

Advance Features of Powerpoint 2016



Let's Warm Up the Brain

Learning Goals

- Powerpoint 2016
- Action Buttons
- Printing the Presentation
- Printing the Presentation

Have you ever seen a presentation with moving text, colourful transitions, videos, and clickable buttons? These features make a presentation more interesting and interactive for the audience. What advanced features do you think can make a simple PowerPoint presentation more attractive and engaging?

Have you ever watched a presentation that includes music, sound effects, or videos? Such presentations are usually more interesting because they combine pictures, sound, and motion to explain ideas in an engaging way. Adding audio and video helps capture the audience's attention and makes information easier to understand. PowerPoint provides simple tools to insert audio recordings, movie clips, and online videos into your slides. It also allows you to use action buttons for easy navigation and print your presentation in different layouts. In this chapter, you will learn how to insert audio and video files, create action buttons, and print presentations effectively.

Inserting Audio and Video Files

PowerPoint allows you to add different types of media, such as audio recordings and video clips, to your presentation. These multimedia elements make your slides more interactive, attractive, and easier to understand. You can insert audio stored on your computer, record your own voice, or add videos from your PC or the Internet. Using audio and video effectively helps keep the audience interested and improves the overall presentation experience.

Inserting an Audio File:

PowerPoint allows you to add background music, sound effects, or voice recordings to your presentation. Audio makes a presentation more engaging and helps explain ideas clearly. You can insert an existing audio file from your computer or record your own voice directly in PowerPoint.

Follow these steps to insert an audio file:



Step 1: Click on the Insert tab. In the Media group, click the Audio button.

Step 2: A drop-down menu appears with the following options:

Audio on My PC

Choose this option to insert an audio file stored on your computer. The Insert Audio dialog box opens. Browse to the location where the audio file is saved, select the desired file, and click the Insert button. The selected audio file is added to the current slide.

Record Audio

This option allows you to record your own voice directly in PowerPoint. When you click Record Audio, the Record Sound dialog box appears. Type a suitable name for your recording and click the Record button to begin recording your voice. After you finish speaking, click the Stop button. You can click the Play button to listen to your recording. If you are satisfied with it, click OK to insert the recording into the slide.

Step 3: After the audio is inserted, a speaker icon appears on the slide. You can drag this icon to any position on the slide. During the presentation, click the Play/Pause button to play or pause the audio. A progress bar also appears, showing how much of the audio has already been played.

Step 4: When you select the audio icon, two contextual tabs Audio Format and Playback appear on the Ribbon. These tabs provide various options to change the appearance of the audio icon and control how the audio plays during the slideshow.

Inserting a Video File

Videos make a presentation more interesting, attractive, and easier to understand. They help explain a topic with moving pictures and sound, making learning more enjoyable. PowerPoint 2016 allows you to insert videos stored on your computer as well as videos available online. Once inserted, the video can be played during the slide show whenever required.

To insert a video file, follow these steps:

Step 1: Click the Insert tab. In the Media group, click the Video command.

Step 2: A drop-down menu appears with the following options: Online Video, Video on My PC

Online Video

The Online Video option allows you to insert videos directly from the Internet without downloading them to your computer. This feature is useful when you want to include educational videos or other online content in your presentation.

Step 3: Click Online Video.

Step 4: The Insert Video dialog box appears. Enter the name or link of the required online video in the search box.

Step 5: Select the desired video and click the Insert button.



The selected video is inserted into the current slide. During the slide show, you can play the video by clicking the Play button.

Video on My PC

If the required video is already saved on your computer, you can insert it using the Video on My PC option. This method is useful because the video can be played even without an Internet connection.

Step 1: Click Video on My PC from the drop-down menu.

Step 2: The Insert Video dialog box appears. Browse to the location where the video file is stored.

Step 3: Select the required video file and click the Insert button.

Step 4: The selected video appears on the slide. You can drag it to any desired position and resize it by using the sizing handles.

Action Buttons

PowerPoint allows you to make a presentation interactive by adding Action Buttons. An Action Button is a ready-made button that performs a specific action when clicked or when the mouse pointer is moved over it. It can be used to move to another slide, open another presentation, open a webpage, run a program, or play a sound.

Action Buttons are especially useful in self-learning presentations, quizzes, educational projects, and presentations where the audience can choose different topics by clicking buttons. These buttons help the presenter navigate through the presentation quickly and smoothly.

To insert an Action Button, follow the steps given below:

Step 1: Open the presentation in which you want to insert an Action Button.

Step 2: Click the Insert tab. Under the Illustrations group, click the Shapes command.

Step 3: The Shapes drop-down gallery appears. Scroll down to the bottom of the gallery. The last category is Action Buttons. It contains different built-in buttons, such as Home, Back, Forward, Beginning, End, Information, Help, Movie, and Sound. Select the Action Button that you want to use.

Step 4: Move the mouse pointer to the slide. Click and drag the mouse to draw the Action Button. Release the mouse button after drawing the required size. As soon as the Action Button is inserted, the Action Settings dialog box appears automatically.

Step 5: Click the Mouse Click tab if it is not already selected. Select Hyperlink to: and choose the required option from the drop-down list.

Step 6: If required, tick the Play Sound option and select a sound from the list. This option plays a sound whenever the Action Button is clicked.

Step 7: Click the OK button.

The Action Button is now ready to use. During the Slide Show, clicking the button performs the selected action.



Printing the Presentation

After creating a presentation, you may want to keep a printed copy for future reference or distribute it among students, teachers, or the audience. PowerPoint 2016 provides different printing options that allow you to print the presentation in various layouts. You can print the entire presentation, selected slides, handouts, notes pages, or only the outline of the presentation. Printing a presentation is useful during classroom teaching, seminars, meetings, and project discussions, as it allows the audience to follow the presentation easily.

Follow these steps to print a presentation.

Step 1: Click the File tab.

Step 2: The Backstage View appears. Click the Print option from the left pane.

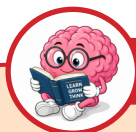
Step 3: The Print window opens. It displays the print settings on the left and the Print Preview on the right. The preview shows how the selected pages will look after printing.

Some commonly used options are:

- **Print All Slides:** Prints every slide in the presentation.
- **Print Selection:** Prints only the selected slides.
- **Print Current Slide:** Prints only the slide currently displayed.
- **Custom Range:** Prints only the slide numbers entered by the user.

Step 4: Select the desired print options.

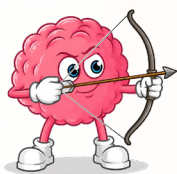
Step 5: Click the Print button.



Quick Recap

- The Print option is available in the File tab.
- The Print Preview shows how the presentation will look after printing.
- You can print all slides, selected slides, or a custom range.
- Full Page Slides print one slide on each page.
- Notes Pages print slides along with speaker notes.
- Outline prints only the text of the presentation.
- Handouts print multiple slides on one page.
- PowerPoint provides Colour, Grayscale, and Pure Black and White printing options.





TEST YOUR SKILLS



A. Choose the correct option

- Which command is used to insert an audio file stored on your computer?
a. Record Audio () b. Audio on My PC ()
c. Screen Recording () d. Online Video ()
- Which tab appears when you select an inserted video on a slide?
a. Design () b. Home ()
c. Video Tools () d. Review ()
- Which Action Button is commonly used to move to the next slide?
a. Home () b. Next ()
c. Help () d. Return ()
- Which print option prints only the slide that is currently open?
a. Print Selection () b. Print Current Slide ()
c. Print All Slides () d. Custom Range ()
- Which feature allows you to add your own voice directly into a presentation?
a. Audio on My PC () b. Online Video ()
c. Record Audio () d. Screen Recording ()

B. Fill in the Blanks

- The _____ tab contains options to control how a video or audio plays.
- The _____ view appears after clicking the File tab.
- The _____ option performs an action when the pointer is moved over an Action Button.
- Click the _____ command to get a paper copy of the presentation.
- A _____ can be inserted from your computer or from the Internet.

C. State True or False

- PowerPoint allows you to record your own voice into a presentation. _____
- Action Buttons can be used only while editing a presentation. _____
- Print Preview helps you check the presentation before printing. _____
- Videos inserted from Internet can be played without an Internet. _____
- Handouts allow multiple slides to be printed on one page. _____



D. Short Answer Type Questions

1. Why is audio added to a presentation?
2. What is the purpose of an Action Button?
3. Name any two print options available in PowerPoint.
4. Why should we check Print Preview before printing?

D. Long Answer Type Questions

1. Explain the steps to insert an audio file from your computer into a PowerPoint presentation.
2. Write the steps to insert an Action Button in a presentation.
3. Explain how to print a presentation using the Print option.

F. Competency-Based Questions

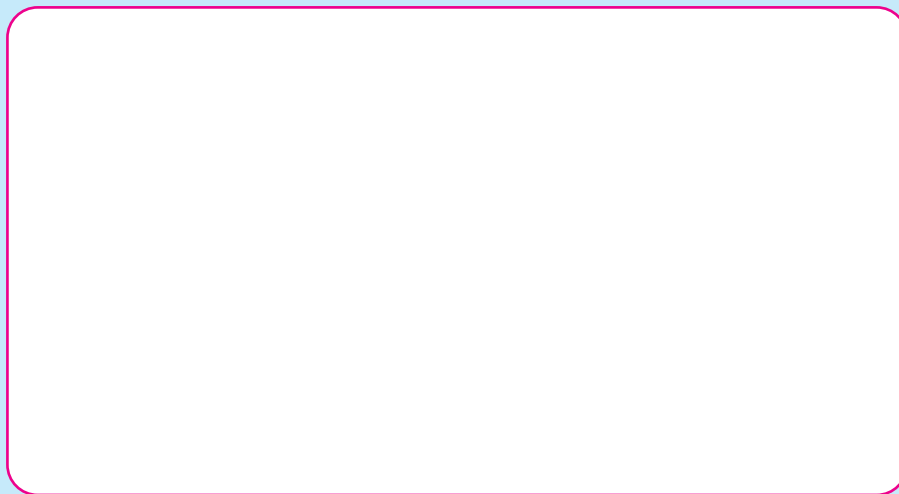
- a) Your teacher asks you to prepare a presentation on "Save Water". You have a video saved on your computer that explains rainwater harvesting. Describe how you will insert this video into your presentation.
- b) You are preparing an interactive quiz in PowerPoint. Students should be able to move to the next question by clicking a button. Which feature will you use? Explain how it will help.



SDG Activity

Work in pairs and create a presentation on "Digital Learning in Schools."

Add at least one audio recording, one video, and Action Buttons for easy navigation.





Practical Exercises

1. Create a PowerPoint presentation on the topic “Our Solar System” using a suitable theme. Prepare separate slides for the Sun, the eight planets, and other important members of the Solar System. Insert suitable pictures on each slide and write a few lines about them. Record a short audio explaining each planet and insert it into the respective slide. Add Action Buttons to move between the slides and print the presentation using the Handouts layout.
2. Prepare a PowerPoint presentation on “Save Environment”. Create different slides on topics such as planting trees, reducing pollution, saving water, and recycling waste. Insert suitable pictures and add a short video related to environmental conservation. Record your own voice explaining the importance of protecting the environment and insert the audio into the presentation. Use Action Buttons to navigate through the slides and print the presentation after checking the Print Preview.
3. Design a PowerPoint presentation on “Famous Monuments of India”. Create separate slides for monuments such as the Taj Mahal, Qutub Minar, India Gate, Charminar, and Gateway of India. Mention their location, the year of construction, the ruler or person associated with them, and one interesting fact. Insert suitable pictures, add an audio narration for each monument, and include an online or offline video related to any one monument. Use Action Buttons to move between the slides and print the presentation as Full Page Slides.

Teacher's Note

- Encourage students to create a multimedia presentation using the features learned in this chapter.
- Assess students based on creativity, correct use of multimedia, and presentation skills.



More on Excel 2016

Learning Goals

- Selecting Cells
- Copy/Move Data
- Merging Cells
- Splitting the Cells
- Formatting Sheets
- Auto Fill
- Worksheet Tab
- Order of Operation
- Using Formulas



Let's Warm Up the Brain

Look at the worksheet carefully and answer the following questions.

	A	B	C	D
1	John	Thomas	Arthur	Micheal
2	210	125	111	109
3	100	200	300	400
4	124	300	234	525

- Which cell contains the highest value in the worksheet?
- What is the value entered in cell C3?
- Which row has the greatest total of all its values?
- Name any two cells that contain values greater than 200.

In the previous class, you learnt the basics of Microsoft Excel, such as creating a workbook, entering data, and saving a workbook. In this chapter, you will learn more about Excel. You will learn how to select and edit cells, copy and move data, insert rows and columns, format worksheets, customise worksheet tabs, use Auto Fill, and perform calculations using formulas. These features help you work more efficiently and create well-organised worksheets.

Selecting Cells in a Worksheet

A worksheet in Microsoft Excel is made up of many rows and columns. The place where a row and a column meet is called a cell. Before performing any task such as entering data, copying information, formatting text, or deleting content, we must first select the required cell or group of cells. Excel provides different ways to select cells depending on the work we want to do. Selecting the correct cells helps us perform operations quickly and accurately.

Single Cell: A single cell is selected when you want to work with only one cell. After selecting the cell, you can enter data, edit its contents, or apply formatting.

Group of Cells or Range: Sometimes we need to work with several cells at the same time. A collection of selected cells is called a cell range. A cell range can be used to copy data, change formatting, apply borders, or perform calculations.

Unrelated or Discontinued Cells: Sometimes the cells that you want to select are not next to each other. These are called discontinuous or non-adjacent cells.

Entire Row: If you want to work with all the cells in a row, you can select the entire row, Move the mouse pointer to the row heading on the left side of the worksheet, and Click the row number. The complete row will be selected.





Entire Column: To work with all the cells in a column, select the entire column, Move the mouse pointer to the column heading at the top of the worksheet, Click the column letter, The complete column will be selected.

Entire Worksheet: If you want to select every cell in the worksheet, you can use the Select All button, Click the Select All button located at the top-left corner of the worksheet where the row and column headings meet, Press Ctrl + A on the keyboard, The entire worksheet will be selected.

Modifying Cell Content

Sometimes, after entering data into a worksheet, we may need to correct a spelling mistake, update information, or make other changes. Editing the existing data in a cell is called modifying cell content. Excel allows us to edit the contents of a cell without deleting the entire data.

Steps to Modify Cell Content

Step 1: Open the worksheet containing the data.

Step 2: Double-click the cell that you want to edit.

Step 3: The insertion point (blinking cursor) appears inside the cell, and Excel enters Edit Mode.

Step 4: Make the required changes.

Step 5: Press the Enter key.

The modified content is displayed in the cell.



Replacing Cell Content

Sometimes, instead of editing the existing text, we may want to replace it completely with new information. In this case, the old content is removed and the new content takes its place.

Steps to Replace Cell Content

Step 1: Click the cell whose content you want to replace.

Step 2: Start typing the new content, The previous content will be replaced automatically.

Step 3: The insertion point (blinking cursor) appears inside the cell, and Excel enters Edit Mode.

Step 4: Press the Enter key.

The new data is stored in the selected cell.

Copying and Moving Data

While working in Excel, we often need to use the same information in more than one place or move data from one location to another. Instead of typing the data again, Excel provides Copy, Cut, and Paste commands.

Cut, Copy and Paste Data

The Copy command creates another copy of the selected data while keeping the original data in its place. The Cut command moves the selected data from one location to another by removing it from its original position. The Paste command inserts the copied or cut data into the selected cell.

Steps to Cut, Copy and Paste Data

Step 1: Select the cell or range of cells that you want to move or copy.

The selected cells will be highlighted.

Step 2: Click the Home tab.

In the Clipboard group, click the required command.

- Click Cut if you want to move the selected data.
- Click Copy if you want to create a duplicate of the selected data.

You can also use the shortcut keys Ctrl + X for Cut and Ctrl + C for Copy.

A moving dotted border appears around the selected cells, indicating that the data has been copied or cut.

Step 3: Click the cell where you want to paste the data.

The selected cell becomes the destination cell.

Step 4: Click the Paste command in the Clipboard group on the Home tab.

You can also press Ctrl + V on the keyboard.

The copied or cut data is pasted into the selected cell.

- If you used Copy, the original data remains in its original location.
- If you used Cut, the data is removed from its original location and moved to the new location.

Column Width and Row Height

Every column has a fixed width and every row has a fixed height. Sometimes the data entered in a cell may be too long to fit within the available space. In such cases, we can increase or decrease the width of a column or the height of a row according to our needs. Changing the column width and row height makes the worksheet neat, readable, and properly arranged. The default column width in Excel is 8.43 characters, and the default row height is 15 points.



Set Column Width/Row Height

Step 1: Select the column(s) or row(s) whose width or height you want to change.

The selected column or row will be highlighted.

Step 2: Click the Home tab.

In the Cells group, click the Format command.

A drop-down menu will appear.

Step 3: Under the Cell Size section, click one of the following options:

- Column Width to change the width of a column.
- Row Height to change the height of a row.

A Column Width or Row Height dialog box will appear.

Step 4: Enter the required value in the dialog box.

Type the desired width or height according to your requirement.

Step 5: Click the OK button.

The selected column or row is resized to the specified width or height.

Automatically Fit Column/Row Content

If the contents of a cell are not fully visible, you can use the AutoFit feature. It automatically adjusts the width of a column or the height of a row according to the contents of the cells.

To automatically fit a column or row, follow these steps:

Step 1: Select the required column(s) or row(s).

Step 2: Move the mouse pointer to the boundary of the selected column heading or row heading.

Step 3: When the pointer changes to a double-headed arrow, double-click the boundary.

The selected column(s) or row(s) are automatically adjusted to fit their contents.

Inserting Rows/Columns

Sometimes, while working on a worksheet, you may need to add more rows or columns to enter additional data. Microsoft Excel allows you to insert new rows and columns without affecting the existing data. When a new row is inserted, the existing rows move downward. Similarly, when a new column is inserted, the existing columns move to the right.

To insert a row or column, follow these steps:

Step 1: Select the row above which you want to insert a new row or the column to the left of which you want to insert a new column.

Step 2: On the Home tab, click the Insert option in the Cells group.

Step 3: Click Insert Sheet Rows to insert a new row or Insert Sheet Columns to insert a new column.

A new row or column is inserted at the selected location.



Merge and Splitting Cells

Sometimes, you may want to display a heading across two or more columns. In such cases, you can merge adjacent cells into a single cell. If required, the merged cell can later be split back into individual cells.

Merge Cells

To merge cells, follow these steps:

Step 1: Select the adjacent cells that you want to merge.

Step 2: On the Home tab, click the Merge & Center option in the Alignment group.

The selected cells are merged into a single cell, and the contents are centred.

Split (Unmerge) Cells

To split a merged cell, follow these steps:

Step 1: Select the merged cell.

Step 2: On the Home tab, click the drop-down arrow beside Merge & Center.

Step 3: Click Unmerge Cells.

The merged cell is split into individual cells.

Formatting Spreadsheets

Formatting changes the appearance of the data in a worksheet without changing its contents. It makes the worksheet more attractive, organised, and easier to read. Microsoft Excel provides various formatting options such as Wrap Text, Number Formatting, Cell Borders, Cell Styles, and Fill Colour.

Wrap Text

Sometimes, the text entered in a cell is too long to fit within the column width. The Wrap Text feature displays the text on multiple lines within the same cell.

To apply Wrap Text, follow these steps:

Step 1: Select the required cell or range of cells.

Step 2: On the Home tab, click the Wrap Text option in the Alignment group.

The text is displayed on multiple lines within the selected cell.

Formatting Numbers

Excel provides different number formats to display numbers in various ways, such as Currency, Percentage, Date, and Time.

To apply number formatting, follow these steps:

Step 1: Select the cell or range of cells containing the numbers.

Step 2: On the Home tab, click the drop-down arrow in the Number group.

Step 3: Select the required number format.

The selected number format is applied to the cells.

Cell Borders

Borders are lines added around cells to improve the appearance of a worksheet and make data easier to read.

To apply cell borders, follow these steps:

Step 1: Select the required cell or range of cells.

Step 2: On the Home tab, click the drop-down arrow beside the Borders option in the Font group.

Step 3: Select the required border style.

The selected border is applied to the cells.

To remove a border, select the cells, click the Borders drop-down arrow, and choose No Border.

Cell Styles

Cell Styles are predefined combinations of formatting options that can be applied to cells with a single click.

To apply a cell style, follow these steps:

Step 1: Select the required cell or range of cells.

Step 2: On the Home tab, click Cell Styles in the Styles group.

Step 3: Select the desired style.



The selected style is applied to the cells.

Filling Colour in a Cell

The Fill Colour feature is used to change the background colour of selected cells. It helps highlight important information and makes the worksheet more attractive.

To apply a fill colour, follow these steps:

Step 1: Select the required cell or range of cells.

Step 2: On the Home tab, click the drop-down arrow beside the Fill Colour option in the Font group.

Step 3: Select the desired colour.

The selected colour is applied to the background of the cells.

Auto Fill

Entering the same type of data repeatedly can be time-consuming. Microsoft Excel provides the Auto Fill feature, which automatically fills a series of numbers, dates, months, days, or other data in consecutive cells. This feature saves time and makes data entry easier.

To use the Auto Fill feature, follow these steps:

Step 1: Enter the starting value or values of the series in one or more cells.

Step 2: Select the cell or cells containing the data.

Step 3: Move the mouse pointer over the Fill Handle (the small square at the lower-right corner of the selected cell). The pointer changes into a black plus (+) sign.

Step 4: Drag the Fill Handle over the required cells and release the mouse button.

Excel automatically fills the selected cells with the appropriate series.

Using Formulas To Calculation

One of the most useful features of Microsoft Excel is its ability to perform calculations automatically. A formula is an expression used to calculate values in a worksheet. Every formula in Excel begins with an equal (=) sign. A formula may contain numbers, cell references, operators, or functions.

To enter a formula, follow these steps:

Step 1: Click the cell where you want to display the result.

Step 2: Type the equal (=) sign.

Step 3: Enter the formula using numbers or cell references. For example, type =A2+B2.

Step 4: Press the Enter key.

The result of the formula is displayed in the selected cell.

Copying a Formula

After creating a formula in a worksheet, you may need to use the same formula for other rows or columns. Instead of entering the formula again, Microsoft Excel allows you to copy it to other cells. This saves time and ensures that calculations are performed accurately. When a formula is copied, Excel automatically changes the cell references according to the new location.

To copy a formula, follow these steps:

Step 1: Select the cell that contains the formula you want to copy.

Step 2: Move the mouse pointer over the Fill Handle (the small square at the lower-right corner of the selected cell). The pointer changes into a black plus (+) sign.

Step 3: Click and drag the Fill Handle over the cells where you want to copy the formula.

Step 4: Release the mouse button.

The formula is copied to the selected cells, and Excel automatically calculates the results by adjusting the cell references.

Order of Operation

A formula in Microsoft Excel may contain more than one mathematical operator. In such cases, Excel follows a fixed sequence to perform the calculations. This sequence is called the Order of Operation. Excel follows the BEDMAS rule while evaluating a formula. According to this rule, calculations inside brackets are performed first, followed by exponents, division, multiplication, addition, and subtraction. If two operators have the same priority, Excel performs the calculations from left to right. Following the correct order of operation ensures that Excel always produces the correct result.

For example: consider the formula $=10+5*2$. Excel first performs the multiplication ($5 \times 2 = 10$) and then adds 10 to obtain the final answer 20.

Similarly, in the formula $=(10+5)*2$, Excel first calculates the value inside the brackets ($10 + 5 = 15$). It then multiplies the result by 2 to obtain the final answer 30.

In the formula $=(20-8)/4+6$, Excel first calculates the expression inside the brackets ($20 - 8 = 12$). It then divides 12 by 4 to get 3 and finally adds 6 to obtain the answer 9.



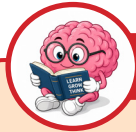


Activity

Enter the following formulas in Excel and write the answers obtained.

- $=15+5*2$
- $=(15+5)*2$
- $=30-12/3$
- $=(30-12)/3$

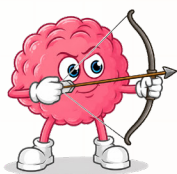
Compare the results and discuss why they are different.



Quick Recap

- Cells, rows, columns, or the entire worksheet can be selected to perform different tasks.
- Data in a worksheet can be modified, copied, moved, or replaced easily.
- The size of rows and columns can be changed manually or by using the AutoFit feature.
- Excel provides various formatting features such as Wrap Text, Borders, Cell Styles, and Fill Colour.
- Worksheet tabs can be renamed, coloured, or deleted to organise a workbook.
- The Auto Fill feature helps fill a series of data automatically.
- Formulas always begin with an equal (=) sign and are used to perform calculations.
- Excel follows the BEDMAS rule while evaluating formulas.





TEST YOUR SKILLS



A. Choose the correct option

- Which command is used to combine two or more adjacent cells into one cell?
a. Wrap Text ☐ b. Merge & Center ☐
c. Cell Styles ☐ d. Auto Fill ☐
- Which feature automatically completes a series of numbers, months, or days?
a. Copy ☐ b. Auto Fill ☐
c. Paste ☐ d. Format Painter ☐
- Every formula in Excel begins with which symbol?
a. + ☐ b. # ☐
c. = ☐ d. % ☐
- Which option is used to change the background colour of a cell?
a. Font Color ☐ b. Fill Color ☐
c. Cell Border ☐ d. Wrap Text ☐
- Excel follows which rule while performing calculations?
a. BODMAS/BEDMAS ☐ b. Alphabet Rule ☐
c. Decimal Rule ☐ d. Rounding Rule ☐

B. Fill in the Blanks

- The _____ feature displays long text on multiple lines in the same cell.
- The small square at the lower-right corner of a selected cell is called _____.
- Cell _____ make tables neat and easy to read.
- We can _____ a worksheet tab to give it a meaningful name.
- Every Excel _____ begins with an equal (=) sign.

C. State True or False

- Auto Fill can be used to fill the names of months automatically. _____
- Cell Borders improve the appearance of a worksheet. _____
- A worksheet tab cannot be renamed. _____
- The Fill Handle can be used to copy formulas. _____
- Excel performs calculations only after the Enter key is pressed. _____



D. Short Answer Type Questions

1. What is the difference between Copy and Cut commands?
2. Why is the Auto Fill feature useful?
3. What is the purpose of merging cells?
4. Name any four number formats available in Excel.

D. Long Answer Type Questions

1. Write the steps to insert a new row in a worksheet.
2. Write the steps to apply Fill Colour to a cell.
3. What is a formula? Explain how to enter a formula in Excel with an example.

F. Competency-Based Questions

- a) Rahul has entered the names of the twelve months manually in his worksheet. His teacher tells him that there is a faster method. Which Excel feature should Rahul use, and how will it help him?
- b) A shopkeeper wants to calculate the total price by adding the values stored in cells B2 and C2. Write the formula he should enter and explain why the formula begins with an equal (=) sign.



Idea Lab

Collect information from the Internet or your computer teacher about any three advanced features of Microsoft Excel, such as Charts, Pivot Tables, Conditional Formatting, or Data Validation.



Practical Exercises

Create a worksheet showing the marks of eight students in five subjects. Then perform the following tasks:

- Rename the worksheet as Class VI Result.
- Insert a new column between English and Mathematics and name it Computer.
- Use Auto Fill to enter roll numbers from 1 to 8.
- Apply Merge & Center to create a heading for the marksheet.
- Apply All Borders and fill the heading with a light colour.
- Calculate the Total Marks of each student using a formula.
- Increase the width of the Student Name column using AutoFit.



Teacher's Note

- *Encourage students to perform all the activities given in the chapter.*
- *Give students sufficient hands-on practice to strengthen their understanding of the concepts*
- *Motivate students to explore the features of the software independently.*
- *Ask students to maintain neatness and accuracy while performing practical work.*



Formulas and Functions in Excel 2016

Learning Goals

- Types of Data
- Formulas in Excel
- Cell Range
- Cell Referencing in Formulas
- References to other worksheets
- Functions



Let's Warm Up the Brain

Suppose you scored 95, 85, and 90 marks in Mathematics, English, and Computer Science, respectively. Open Microsoft Excel and enter these marks in a worksheet. Find the total marks, average marks, and percentage using Excel formulas.

Microsoft Excel is one of the most popular spreadsheet programs used to organize, calculate, and analyze data. It allows us to perform mathematical calculations quickly and accurately using formulas and functions. Excel is widely used in schools, offices, businesses, banks, hospitals, and many other organizations for maintaining records and preparing reports. In addition to calculations, Excel provides various tools to arrange information in rows and columns, create charts, sort and filter data, and perform automatic calculations. Since formulas update their results whenever the data changes, Excel saves both time and effort.

In this chapter, you will learn about the different types of data in Excel, methods of entering formulas, cell ranges, different types of cell references, references to other worksheets, and commonly used Excel functions.

Types of Data in Excel

A worksheet is made up of rows and columns. The point where a row and a column intersect is called a cell. Every cell in Excel can store different kinds of information known as data. Excel mainly accepts three types of data: Labels, Values (Numbers), and Formulas. Each type serves a different purpose in a worksheet.

Labels: Labels are text entries that describe the information in a worksheet. They are generally used as headings, titles, names, or descriptions to make the worksheet easy to understand. Labels do not participate in mathematical calculations.

Values: Values are numerical data on which Excel performs calculations. They include numbers from 0 to 9 and may also contain decimal points, currency symbols, percentages, or negative numbers.



Formula: A formula is a mathematical expression used to perform calculations on data entered in one or more cells. Every formula in Excel always begins with an equal (=) sign. Excel automatically calculates the result and displays it in the selected cell. Example: = A2 + B2

Different ways to Enter a Formulas

There are mainly two ways to enter a formula in an Excel worksheet:

Typing the Formula in Cell

In this method, the formula is entered directly into the selected cell after typing an equal (=) sign. After entering the formula, press the Enter key. Excel calculates the formula and displays the result in the same cell.

Typing the Formula in the Formula Bar

Another way to enter a formula is by using the Formula Bar. First, select the required cell and then type the formula in the Formula Bar after the equal (=) sign. Press the Enter key to calculate the formula. The result appears in the selected cell.

Understanding Cell Range

A cell range is a group of two or more selected cells in a worksheet. Cell ranges make it easier to perform calculations on multiple cells at the same time instead of selecting each cell individually. When a range is selected, Excel highlights all the cells included in that range. A cell range is represented by the address of the first cell, followed by a colon (:) and the last cell. For example, if the selected cells start from B2 and end at E3, the range is written as: B2:E3.

Selecting a Range in Worksheet

By Using the Mouse

This is the easiest and most commonly used method for selecting adjacent cells.

Step 1: Click the first cell from where you want to begin the selection.

Step 2: Press and hold the left mouse button.

Step 3: Drag the mouse across the worksheet until all the required cells are selected.

Step 4: Release the mouse button.

Step 5: The selected cells are highlighted, indicating that the cell range has been selected successfully.

By Using the Shift Key

The Shift key can also be used to select a continuous range of cells without dragging the mouse.

Step 1: Click the first cell of the required range.

Step 2: Press and hold the Shift key on the keyboard.



Step 3: While holding the Shift key, click the last cell of the range.

Step 4: Release the Shift key.

Step 5: Excel automatically selects all the cells between the first and the last selected cell.

Naming a Cell Range

A named range can represent a single cell or a group of cells. Once a name has been assigned, it can be used anywhere in the workbook instead of the cell reference. For example, instead of writing =SUM(B3:B10), we can assign the name Sales to the range and write =SUM(Sales).

Using named ranges also reduces the chances of selecting the wrong cells while creating formulas.

Steps to Name a Cell Range

Step 1: Select the cell or range of cells that you want to assign a name.

Step 2: Click the Formulas tab on the Ribbon.

Step 3: In the Defined Names group, click the Define Name command.

Step 4: The New Name dialog box appears.

Step 5: Type a suitable name for the selected range in the Name box.

Step 6: Click the OK button.

Step 7: The assigned name is stored and can now be used in place of the cell address in formulas.

Cell Referencing in Formulas

A cell reference is the address of a cell that is used in a formula. Instead of entering actual values into a formula, Excel allows us to use the addresses of cells containing those values. This makes formulas dynamic because the result changes automatically whenever the data in the referenced cells changes. Cell references are very useful while working with large worksheets. They eliminate the need to rewrite formulas every time the values change.

For example, if cell A1 contains 10 and cell A2 contains the formula: =A1*2

The result displayed in A2 will be 20. If the value in A1 is changed from 10 to 15, Excel automatically recalculates the formula, and the value in A2 becomes 30.

Relative References

A Relative Reference changes automatically when a formula is copied from one cell to another. It is the default type of cell reference used in Excel. When a formula containing a relative reference is copied to another location, Excel adjusts the row and column references according to the new position. This feature is useful when performing similar calculations for multiple rows or columns.

Example: Suppose cell D3 contains the formula: =B3*C3. If this formula is copied to cell D4, Excel automatically changes it to: =B4*C4. The references change according to the new row.

Steps to Use Relative References

Step 1: Enter the required data into the worksheet.

Step 2: Select the cell where you want to perform the calculation.

Step 3: Type the required formula using cell references.

Step 4: Press the Enter key to calculate the result.

Step 5: Select the cell containing the formula.

Step 6: Copy the formula using the Copy command or the Fill Handle.

Step 7: Paste the formula into another cell or drag the Fill Handle downward.

Step 8: Excel automatically changes the referenced cell addresses according to the new location.

Absolute References

An Absolute Reference keeps the row number and column letter fixed when a formula is copied to another cell. It is represented by placing a Dollar (\$) symbol before both the column letter and the row number. Absolute references are mainly used when one particular cell value, such as a tax rate, discount percentage, or commission rate, must remain constant in every calculation.

Example: If the discount percentage is stored in cell D3, the formula can be written as: =((B3*C3)-((B3*C3)*(\$D\$3))). When this formula is copied to another row, the reference \$D\$3 remains unchanged.

Mixed References

A Mixed Reference is a combination of a relative reference and an absolute reference. In this type of reference, either the row or the column remains fixed while the other changes.

Mixed references are useful when copying formulas across rows or columns where only one part of the cell reference should remain constant.

There are two types of mixed references:

\$A1: The column is fixed, but the row changes.

A\$1: The row is fixed, but the column changes.

References to other Worksheets

Excel allows us to use data stored in different worksheets of the same workbook. This feature is called Referencing Other Worksheets. It enables users to create formulas using values from multiple worksheets without copying the data. When a reference is made to another worksheet, Excel places an exclamation (!) mark between the worksheet name and the cell



address.

Functions

A function is a predefined formula in Microsoft Excel that performs a specific calculation automatically. Instead of writing long and complex formulas, we can use built-in functions to perform calculations quickly and accurately.

Every function accepts one or more arguments. An argument is the value or cell reference supplied to the function for calculation. Arguments are always written inside parentheses (). They can be numbers, text, cell references, or cell ranges.

Rules for Using Functions

- While using functions in Excel, the following rules should always be followed:
- Every function must begin with an equal (=) sign.
- The function name must be a valid Excel function name.
- The function name must be followed by opening and closing parentheses ().
- Arguments should be written inside the parentheses.
- Multiple arguments should be separated by commas or by using a cell range.
- Cell references can also be used instead of directly entering values.
- After typing the function, press the Enter key to display the result.

Categories of Functions

Excel provides many categories of functions to perform different types of calculations. Some of the commonly used categories of functions are given below:

Mathematical Functions: are used to perform arithmetic calculations on numbers. These functions can add, multiply, divide, round numbers, find the square root, calculate powers, and perform many other mathematical operations.

Text Functions: used to perform different operations on text strings. They help in joining text, finding the length of a string, extracting characters from the left or right side, and converting text into uppercase or lowercase letters.

Logical Functions: used to test a condition and return a result based on whether the condition is TRUE or FALSE. These functions are commonly used to make decisions in a worksheet.

Statistical Functions: used to perform calculations on a group of numbers. They help in finding the maximum value, minimum value, average, count, and other statistical information from a range of data.

Date and Time Functions: used to work with dates and time values in Excel. They can display the current date and time or extract the day, month, year, hour, and minute from a date or time value.

Mathematical Functions

Functions	Purpose	Example
SUM(range)	It adds the values in a range of cells.	Input: =SUM(5,10,15) Output: 30
PRODUCT(range)	It multiplies the values in a range of cells.	Input: =PRODUCT(3,4,5) Output: 60
MOD(number, divisor)	It returns the remainder after a number is divided by the divisor.	Input: =MOD(14,5) Output: 4
SQRT(number)	It returns the square root of the given number.	Input: =SQRT(144) Output: 12
INT(number)	It rounds off a number to an integer value.	Input: =INT(18.95) Output: 18
POWER(number,power)	It returns the result of a number raised to a specified power.	Input: =POWER(4,3) Output: 64
ROUND(number,num_digits)	It rounds a number to the specified number of digits.	Input: =ROUND(18.678,2) Output: 18.68
ABS(number)	It returns the absolute value of a given number.	Input: =ABS(-40) Output: 40

Perform the following steps to use the mathematical functions:

Step 1: Type the required data in the Excel worksheet.

Step 2: To calculate the amount of the first item, type the =PRODUCT(B3:C3) function in the required cell.

Step 3: Press the Enter key.

When you press the Enter key, the calculated amount appears in the selected cell. Using the Auto Fill feature, calculate the amount for the remaining items.

Step 4: To calculate the subtotal, type the =SUM(D3:D7) function in the required cell.

Step 5: Press the Enter key.

When you press the Enter key, the subtotal appears in the selected cell.

Step 6: Calculate the GST by using the formula =(D8*5)/100.



Step 7: Calculate the Invoice Amount by using the =SUM(D8:D9) function. The final output appears in the worksheet.

Text Functions

Functions	Purpose	Example
CONCATENATE(text1,text2)	It joins together two or more text strings.	Input: = CONCATENATE ("John", " Doe") Output: John Doe
LEFT(text,num_chars)	It returns the specified number of characters from the left side of a text string.	Input: =LEFT("JOHN", 3) Output: JOH
RIGHT(text,num_chars)	It returns the specified number of characters from the right side of a text string.	Input: =RIGHT("JOHN", 2) Output: HN
LEN(text)	It returns the length of the text string.	Input: =LEN("JOHN") Output: 4
UPPER(text)	It converts the text string into uppercase.	Input: =UPPER("John") Output: JOHN
LOWER(text)	It converts the text string into lowercase.	Input: =LOWER("JOHN") Output: john

Perform the following steps to use the text functions:

Step 1: Type the required data in the Excel worksheet.

Step 2: Type the =CONCATENATE(A3,B3) function in the required cell to join the first name and last name.

Step 3: Press the Enter key.

Logical Functions

Functions	Purpose	Example
IF(condition, value_if_true, value_if_false)	It checks whether a condition is true or false and returns the specified result.	Input: =IF(75>=40,"Pass","Fail") Output: Pass



Statistical Functions

Functions	Purpose	Example
MAX(range)	It returns the largest value in a range of cells.	Input: =MAX(18,45,27,60) Output: 60
MIN(range)	It returns the smallest value in a range of cells.	Input: =MIN(18,45,27,60) Output: 18
AVERAGE(range)	It calculates the average of the selected values.	Input: =AVERAGE(20,40,60) Output: 40
COUNT(range)	It counts the number of cells containing numeric values.	Input: =COUNT(10,25,35,40) Output: 4

Perform the following steps to use the text functions:

Step 1: Enter the marks of students in an Excel worksheet.

Step 2: Select the cell where you want to display the highest marks.

Step 3: Type the formula =MAX(B3:F3).

Step 4: Press the Enter Key

The highest marks obtained by the student are displayed in the selected cell.

Step 5: Use the Auto Fill feature to calculate the highest marks for the remaining students.

Step 6: Similarly, use the MIN(), AVERAGE(), and COUNT() functions to find the lowest marks, average marks, and the number of numeric entries.

Date and Time Functions

Functions	Purpose	Example
TODAY()	It returns the current system date.	Input: =TODAY() Output: 23/07/2024
NOW()	It returns the current system date and time.	Input: =NOW() Output: 23/07/2024 11:24
DAY(date)	It returns the day from a date.	Input: =DAY(TODAY()) Output: 23



MONTH(date)	It returns the month from a date.	Input: =MONTH(TODAY()) Output: 7
YEAR(date)	It returns the year from a date.	Input: =YEAR(TODAY()) Output: 2024
HOUR(time)	It returns the hour from a time value.	Input: =HOUR(NOW()) Output: 11
MINUTE(time)	It returns the minute from a time value.	Input: =MINUTE(NOW()) Output: 24

Perform the following steps to use the TODAY() function:

Step 1: Select the cell in which you want to display the current date.

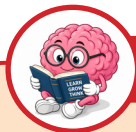
Step 2: Type the formula =TODAY().

Step 3: Press the Enter key.

The current system date appears in the selected cell.

Step 4: Save the workbook.

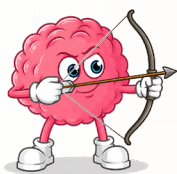
Step 5: Open the workbook on another day to observe that the displayed date is updated automatically.



Quick Recap

- Excel uses formulas and functions to perform calculations.
- A cell range is a group of selected cells.
- Excel supports Relative, Absolute, and Mixed cell references.
- A function is a predefined formula in Excel.
- Functions are grouped into different categories according to their purpose.
- SUM(), AVERAGE(), and IF() are commonly used Excel functions.





TEST YOUR SKILLS



A. Choose the correct option

- Which of the following is an Absolute cell reference?
a. A5 ☐ b. \$B\$4 ☐
c. B\$4 ☐ d. \$B4 ☐
- Which function is used to count the number of numeric values in a range?
a. COUNT() ☐ b. SUM() ☐
c. LEN() ☐ d. MAX() ☐
- Which function converts text into uppercase letters?
a. LOWER() ☐ b. UPPER() ☐
c. LEFT() ☐ d. RIGHT() ☐
- Which symbol separates the worksheet name from the cell address in a worksheet reference?
a. : ☐ b. ! ☐
c. # ☐ d. \$ ☐
- Which function returns the current system date and time?
a. TODAY() ☐ b. NOW() ☐
c. DAY() ☐ d. MONTH() ☐

B. Fill in the Blanks

- A group of selected cells is called a _____.
- The _____ function joins two or more text strings.
- _____ is the default type of cell reference in Excel.
- The _____ function returns the current system date and time.
- The _____ function is used to add values in a range.

C. State True or False

- Every Excel formula begins with an equal (=) sign. _____
- The PRODUCT() function adds the values in a range. _____
- A mixed reference can lock either a row or a column. _____
- The LEN() function returns the number of characters in a text string. _____
- The TODAY() function displays both the current date and time. _____



D. Short Answer Type Questions

1. What is a formula in Excel?
2. Name any two Text functions.
3. What is a cell range?
4. What is the purpose of the IF() function?

D. Long Answer Type Questions

1. Explain the three types of cell references with suitable examples.
2. Describe the steps to assign a name to a cell range.
3. Explain any four rules for using functions in Excel.

F. Competency-Based Questions

- a) Riya has prepared a worksheet containing the marks of students in five subjects. She wants Excel to calculate the average marks automatically whenever any subject marks are changed. Which Excel feature should she use? Explain your answer.
- b) A shopkeeper wants to show the monthly sales of his shop from January to December. Which chart would be the most suitable and why?



Practical Exercises

Create a Student Mark Sheet in Microsoft Excel.

- Enter the names of five students.
- Enter marks obtained in five subjects.
- Use the SUM() function to calculate the total marks.
- Use the AVERAGE() function to calculate the average marks.
- Use the MAX() and MIN() functions to find the highest and lowest marks.
- Use the IF() function to display "Pass" if the average is 40 or above; otherwise display "Fail".

Teacher's Note

- Encourage students to perform calculations using SUM(), AVERAGE(), MAX(), MIN(), IF(), and other commonly used functions.
- Guide students to create worksheets such as mark sheets, invoices, and expense records to develop practical spreadsheet skills.



Introduction to Animate 2024



Let's Warm Up the Brain

Learning Goals

- Create a Motion Tween
- Create a shape tween
- Create a classic tween
- Use masking effects
- Work with layers
- Format text in Animate

Have you ever wondered how cartoons, animated movies, and video games bring characters and objects to life even though they are made from a series of still images? Imagine that you are creating your own animation in Animate 2024. Which object would you choose to animate—a bouncing ball, a flying bird, a racing car, or something entirely different? What type of movement, colour changes, or special effects would you add to make your animation look realistic and interesting? Explain your ideas.

Adobe Animate 2024 is a powerful animation and vector graphics software developed by Adobe. It allows users to create interactive animations, digital artwork, banners, presentations, educational content, and multimedia projects. With its easy-to-use drawing tools, timeline, symbols, and animation features, Adobe Animate helps beginners as well as professionals design creative and engaging animations. In this chapter, you will learn how to start Adobe Animate 2024, create and save a document, understand the components of its window, draw shapes, apply gradient fills, and create symbols for building animation projects.

Starting Adobe Animate 2024

Before creating animations, you must have Adobe Animate 2024 installed on your computer. It is a powerful application used to design vector graphics, interactive content, cartoons, games, advertisements, and web animations. Once the software is installed, you can launch it and begin creating exciting animation projects.

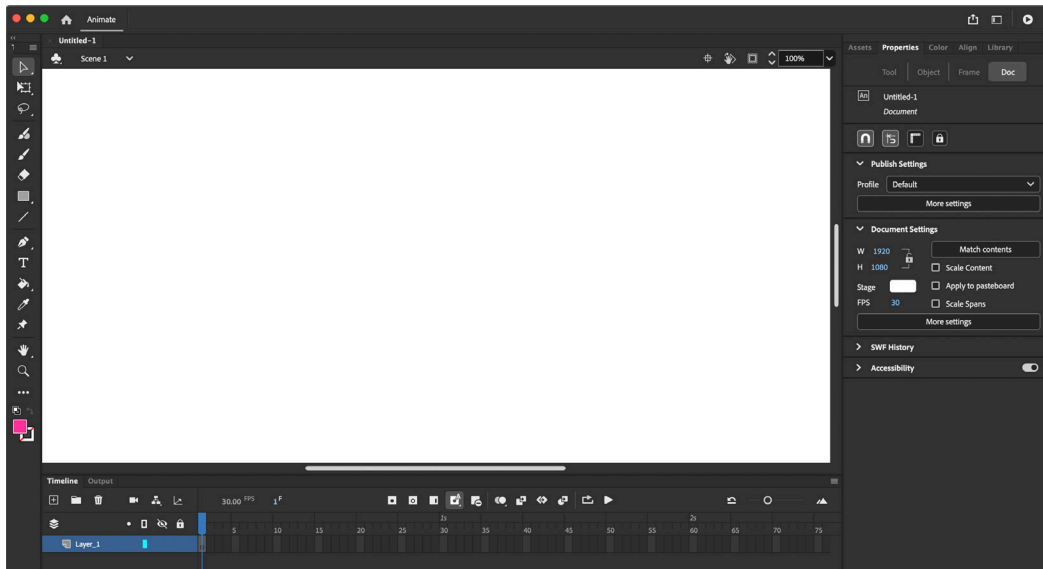
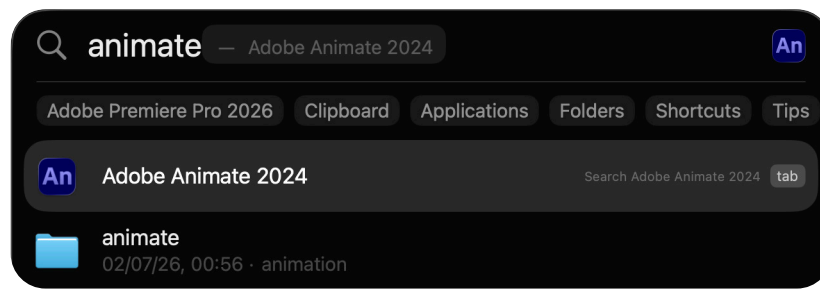
To start Adobe Animate 2024, follow these steps:

Step 1: Click the Start button on the taskbar or press the Windows key on the keyboard.

Step 2: Type Adobe Animate 2024 in the search box. When the application appears in the search results, click on it.

Step 3: The Welcome Screen of Adobe Animate 2024 opens. From here, you can create a new project, open an existing project, or access recently used files.





Creating a Document in Animate 2024

Every animation project begins with a new document. A document provides the workspace where you create drawings, add objects, and build animations. Before you start working, you can choose the document type and adjust settings according to your project requirements.

Follow these steps to create a new document:

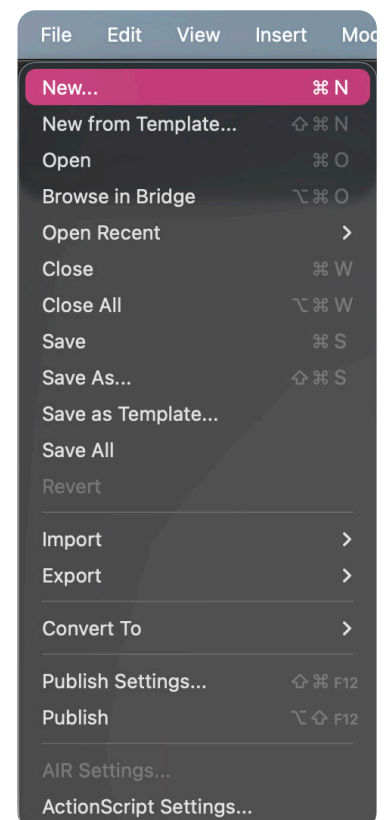
Step 1: Open Adobe Animate 2024.

Step 2: Click File then New, or click the New File button available on the Welcome Screen.

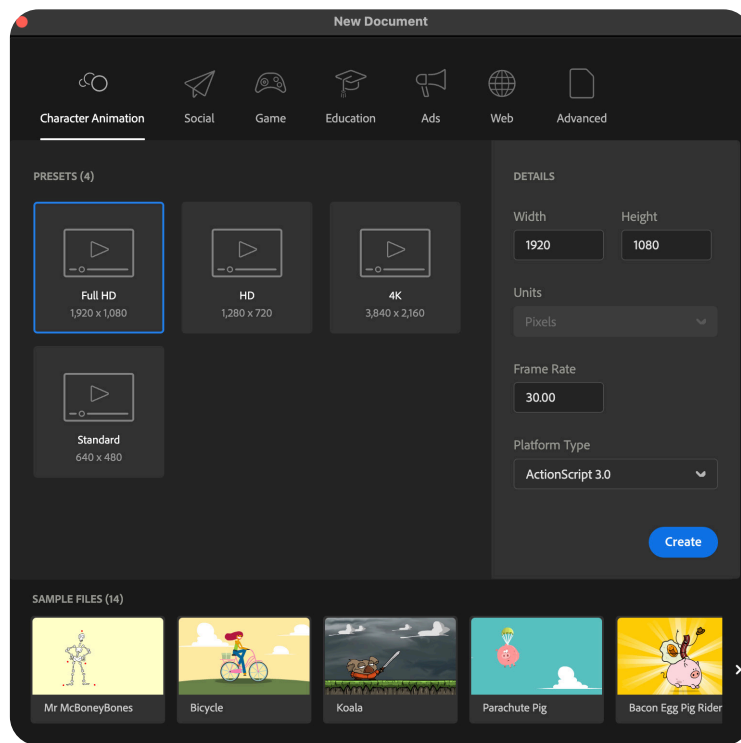
Step 3: The New Document dialog box appears. Under the Character Animation category, select the required preset. For beginners, the HD 1280 × 720 preset is a suitable choice.

Step 4: Set the Width and Height of the Stage according to your project. The default size is 1280 × 720 pixels, which is commonly used for HD animations.

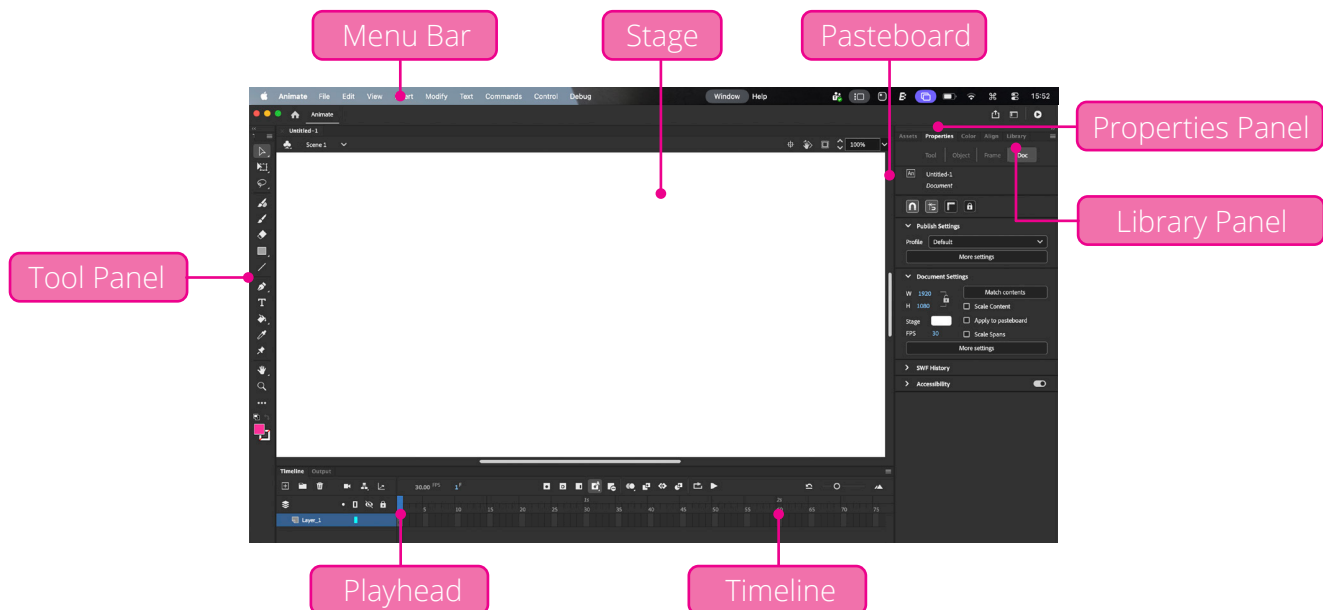
Step 5: Set the Frame Rate to 30 frames per second (fps). The frame rate controls how smoothly an animation plays.



Step 6: Click the Create button. A new Animate document opens and is ready for use.



Components of the Animate 2024 Window



Stage: The Stage is the main working area in Adobe Animate 2024 where we create our drawings and animations. It is a rectangular area in the centre of the Animate window. All the graphics, text, buttons, images, and other objects that are to appear in the final animation are placed on the Stage.



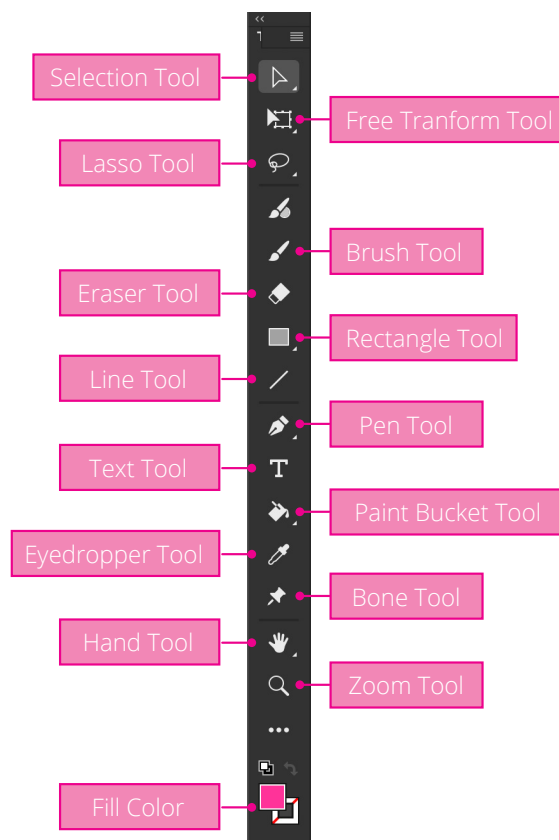
The Stage acts like a digital drawing sheet where we can draw shapes, insert symbols, and arrange different objects. The default size of the HD Stage is 1280 × 720 pixels, but it can be changed according to the requirements of the project. Objects placed outside the Stage are not visible in the final animation.

Timeline: The Timeline is located at the bottom of the Animate window. It controls the timing and sequence of an animation. The Timeline is divided into small boxes called frames, which help us arrange different stages of an animation.

A Playhead moves across the Timeline to show the current frame being viewed or edited. By adding and arranging frames in the Timeline, we can make objects move smoothly and create interesting animations.

Tools Panel: The Tools Panel contains different tools used to draw, select, edit, colour, and modify objects on the Stage. It is usually located on the left side of the Animate window and provides all the basic tools required to create an animation.

Some tools are used for drawing shapes and lines, while others are used for adding text, filling colours, erasing unwanted parts, or selecting objects. By choosing the appropriate tool, we can easily create and edit graphics according to our needs.



The following table gives a brief introduction of tools available in Animate:

Tool Name	Function
Selection	Selects objects on the Stage before they can be edited or modified.
Free Transform	Moves, rotates, scales, skews, and resizes selected objects.
Lasso	Selects irregular portions of objects using freehand selection.
Pen	Draws precise straight lines and smooth curves using anchor points.
Text	Creates and edits text on the Stage.
Line	Draws straight lines of different lengths and colours.
Rectangle	Draws rectangles and squares. Hold Shift to draw a perfect square.

Brush	Creates freehand drawings using brush strokes
Paint Bucket	Fills enclosed areas with the selected colour or gradient.
Eyedropper	Picks colours and styles from existing objects and applies them elsewhere.
Eraser	Removes lines, fills, or parts of drawings.
Hand	Moves the view of the Stage without moving the objects.
Zoom	Zooms in to enlarge the view or zooms out to see more of the workspace.
Assert Wrap	Creates flexible movement by adding control points to an object, making animation easier.

Properties Panel: The Properties Panel is located on the right side of the Animate window. It displays the properties and settings of the selected object or tool. The options available in this panel change automatically depending on what is selected. Using the Properties Panel, you can modify the size, position, colour, stroke, fill, text formatting, and other attributes of an object.

Library Panel: The Library Panel stores all the reusable items used in an Animate project, such as symbols, images, audio files, video files, and buttons. It helps organise project resources in one place and allows you to reuse them whenever required. Objects stored in the Library can be easily dragged onto the Stage.

Menu Bar: The Menu Bar is located at the top of the Animate window. It contains various menus such as File, Edit, View, Insert, Modify, Text, Commands, Control, Window, and Help. These menus provide access to different commands and features that help create, edit, save, and manage animation projects efficiently.

Saving a Document in Animate 2024

After creating an animation project, it is important to save your work regularly. Saving a document protects your project from accidental loss due to power failure or system errors. Adobe Animate saves files with the .fla extension, which stores all the editable contents of your project.

Follow these steps to save a document:

Step 1: Click the File menu on the Menu Bar.

Step 2: Select the Save option. If you are saving the file for the first time, choose Save As.

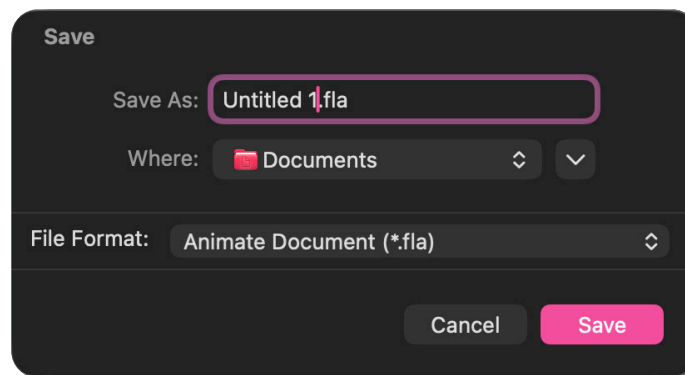
Step 3: The Save As dialog box appears. Browse where you want to save your project.

Step 4: Type a suitable name for the file in the File Name box.

Step 5: Ensure that the file type is Adobe Animate Document (*.fla).

Step 6: Click the Save button.





Creating Shapes in Animate 2024

Shapes are the basic building blocks of every animation. Adobe Animate provides various drawing tools to create rectangles, circles, polygons, lines, and other objects. Every shape consists of two parts:

Fill: The colour inside the shape.

Stroke: The outline or border of the shape.

Both the fill colour and stroke colour can be changed at any time.

Let us learn how to draw a rectangle.

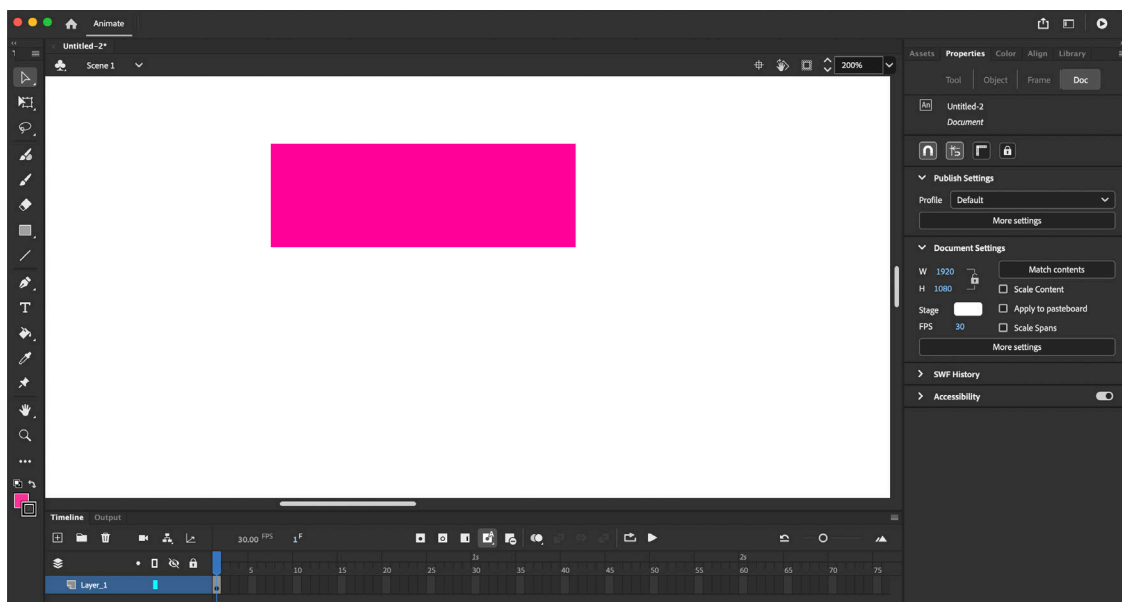
Step 1: Select the Rectangle Tool from the Tools Panel.

Step 2: Choose the desired Fill Colour and Stroke Colour from the colour controls in the Tools Panel.

Step 3: Click and drag the mouse on the Stage to draw a rectangle.

Step 4: Release the mouse button when the rectangle reaches the required size.

Step 5: To draw a perfect square, hold down the Shift key while dragging.



The thickness of the outline is called the Stroke Width. Increasing the stroke width makes the outline thicker, while decreasing it makes the outline thinner.

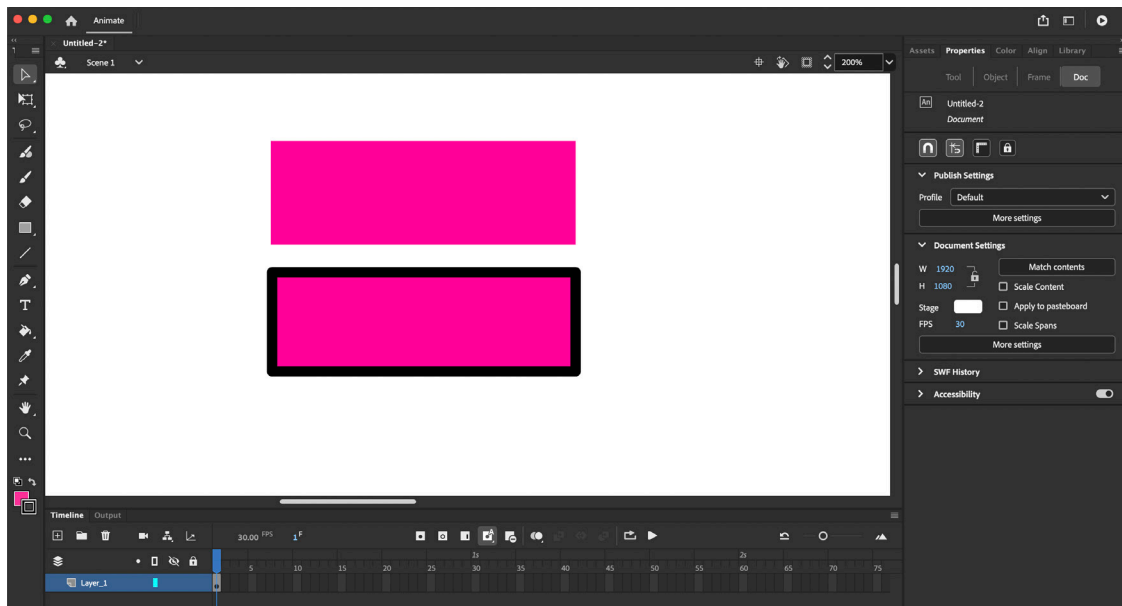
Step 1: Select the Selection Tool.

Step 2: Click the rectangle to select it.

Step 3: Open the Properties Panel.

Step 4: Locate the Stroke Width option.

Step 5: Choose the required width. The outline of the rectangle changes immediately.



Gradient Fill

A Gradient Fill is a special type of fill that smoothly blends two or more colours into one another. It adds depth, shading, and a realistic look to objects. Adobe Animate 2024 provides two types of gradient fills—Linear Gradient, which blends colours in a straight line, and Radial Gradient, which blends colours in a circular pattern. Gradient fills can be applied to shapes, text, and other objects to make them more attractive.

To apply a gradient fill, follow these steps:

Step 1: Select the object to which you want to apply the gradient fill.

Step 2: In the Properties Panel, click the Fill Colour box.

Step 3: From the Fill options, select Linear Gradient or Radial Gradient.

Step 4: The selected object is filled with the default gradient. To customise it, click on the Gradient Colour Bar.

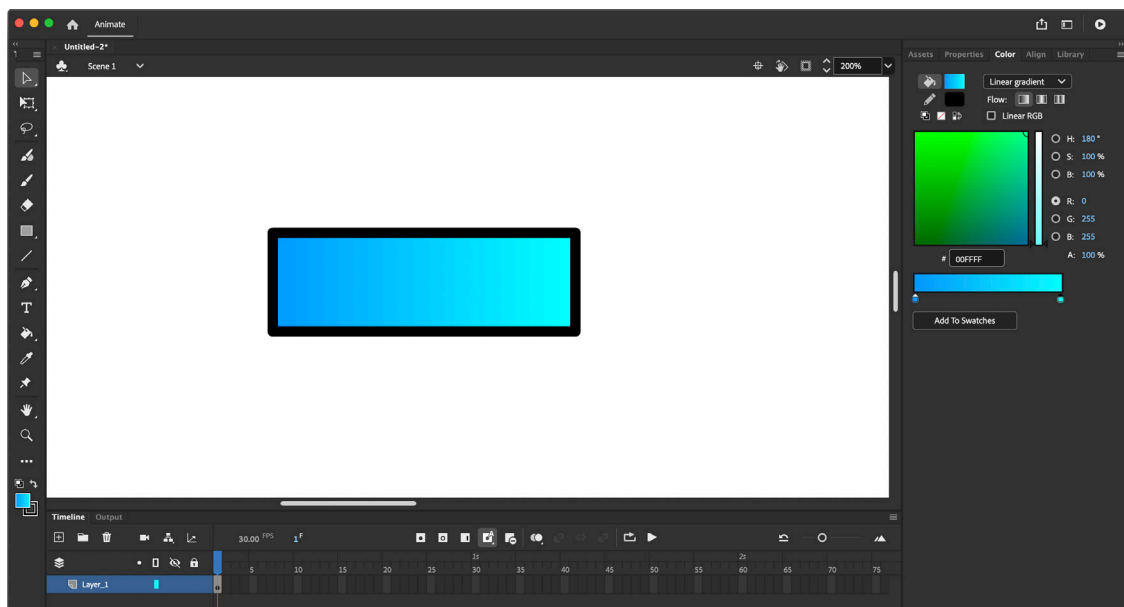
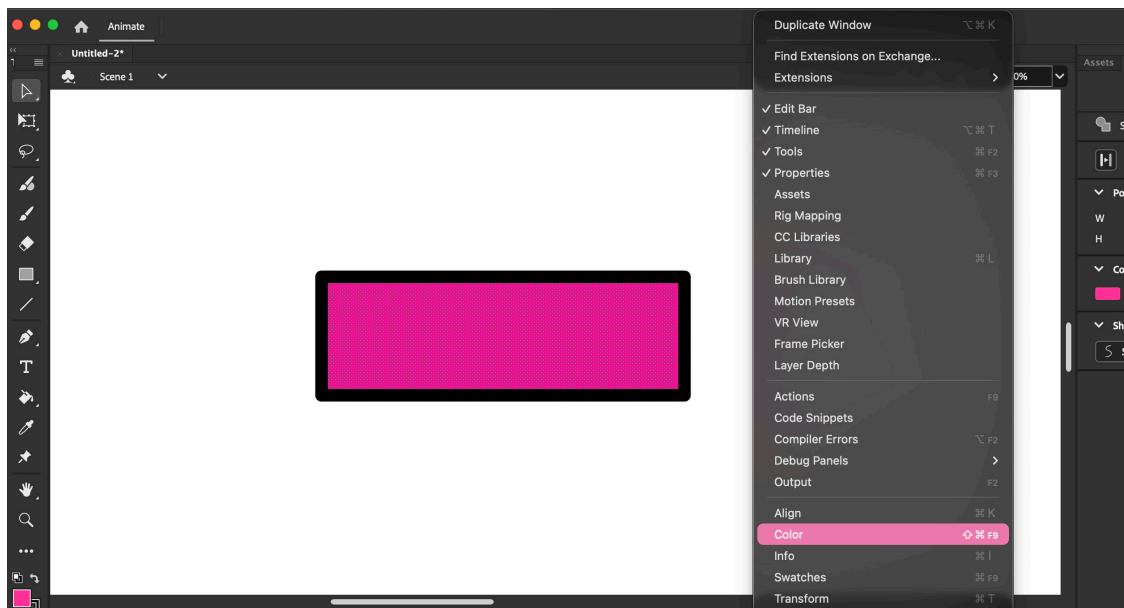
Step 5: Click on a colour stop to change its colour. To add another colour, click below the gradient bar to insert a new colour stop. To remove a colour stop, drag it away from the gradient bar.

Step 6: Adjust the angle for a Linear Gradient or the centre and size for a Radial Gradient until



you get the desired effect.

Step 7: Click anywhere outside the panel to apply the changes. The object now displays the customised gradient fill.



Tech Fact

Gradients are widely used in animations, logos, game graphics, and web design because they make objects appear more colourful and realistic. Adobe Animate mainly provides two types of gradients: **Linear** and **Radial** Gradients.



Creating a Symbol in Animate 2024

A Symbol is a reusable object in Adobe Animate that can be used multiple times in a project without creating it again. Symbols help reduce the file size and make editing easier. Whenever a symbol is modified, all its copies in the project are updated automatically. Symbols are stored in the Library Panel, from where they can be dragged onto the Stage whenever required. When a symbol is placed on the Stage, it is called an Instance of that symbol.

Adobe Animate provides three types of symbols:

Movie Clip Symbol	Used for creating independent animations that have their own Timeline and can include advanced animation effects.
Button Clip Symbol	Used for creating interactive buttons that respond to mouse actions such as clicking or hovering.
Graphic Symbol	Used for creating drawings, illustrations, and simple animations that play along with the Timeline.

To create a symbol, follow these steps:

Step 1: Create a new document or open an existing Animate project.

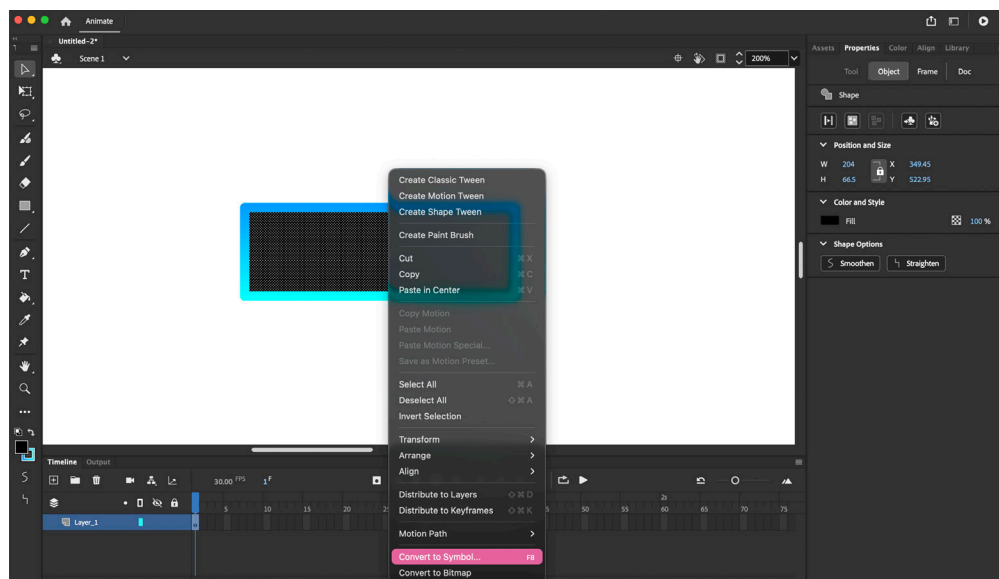
Step 2: Draw a shape on the Stage using any drawing tool, such as the Rectangle Tool, Oval Tool, or Brush Tool.

Step 3: Use the Fill Colour and Stroke Colour options in the Properties Panel to give the object a suitable appearance.

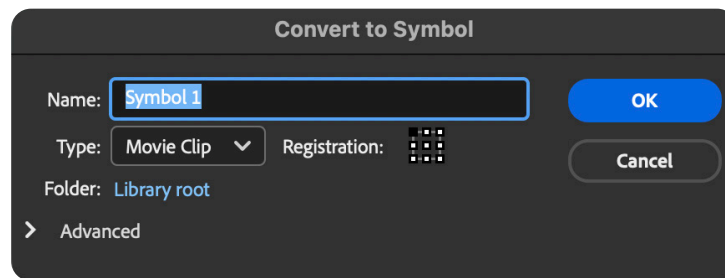
Step 4: Select the object using the Selection Tool.

Step 5: Click the Modify menu on the Menu Bar.

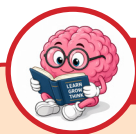
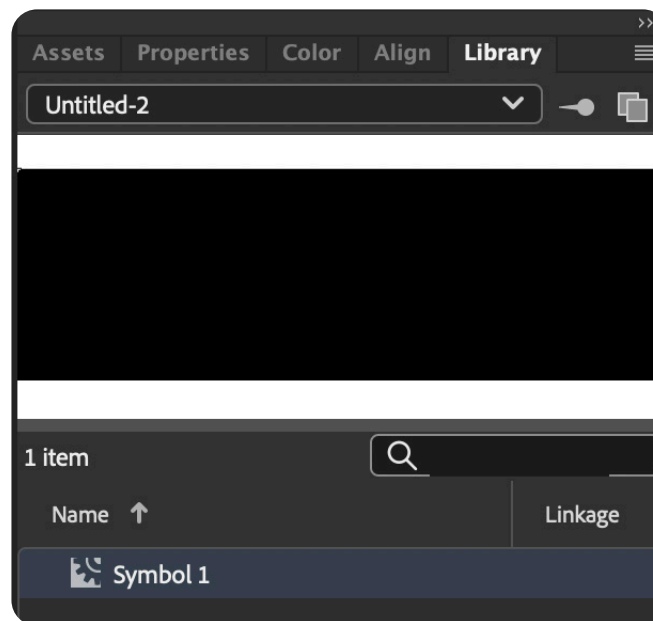
Step 6: Choose Convert to Symbol. Alternatively, you can press the F8 key on the keyboard to open the Convert to Symbol dialog box.



Step 7: The Convert to Symbol dialog box appears. Enter a suitable Name for the symbol. Select the required Symbol Type (Graphic, Button, or Movie Clip). Choose the Registration Point, which determines the reference point of the symbol.



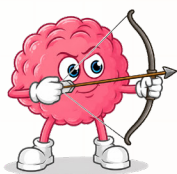
Step 8: Click the OK button. The selected object is converted into a symbol and is automatically stored in the Library Panel.



Quick Recap

- Adobe Animate 2024 is used to create interactive animations and multimedia content.
- The Stage is the main area where objects and animations are created.
- The Timeline controls the sequence and timing of an animation.
- Gradient Fill blends two or more colours smoothly in an object.
- A Symbol is a reusable object stored in the Library Panel.
- Adobe Animate provides Graphic, Button, and Movie Clip symbols.





TEST YOUR SKILLS



A. Choose the correct option

- Which file extension is used to save an Adobe Animate project?
a. .pptx () b. .fiv ()
c. .fla () d. .psd ()
- Which panel stores reusable objects such as symbols and graphics?
a. Timeline Panel () b. Library Panel ()
c. Tools Panel () d. Properties Panel ()
- Which tool is used to fill an enclosed shape with colour?
a. Brush Tool () b. Paint Bucket Tool ()
c. Pen Tool () d. Eraser Tool ()
- Which of the following is NOT a type of symbol in Adobe Animate?
a. Graphic Symbol () b. Button Symbol ()
c. Movie Clip Symbol () d. Layer Symbol ()

B. Fill in the Blanks

Word bank: Stage, Symbol, Library, Gradient, Timeline

- The main working area in Adobe Animate is called the _____.
- A reusable object in Animate is known as a _____.
- All created symbols are stored in the _____ Panel.
- A smooth transition between two or more colours is called a _____ Fill.
- The _____ controls the sequence and timing of an animation.

C. Short Answer Type Questions

- What is the use of the Properties Panel?
- What is a Gradient Fill?
- Name any two components of the Animate 2024 window.

D. Long Answer Type Questions

- Explain the functions of the following tools:
i. Selection Tool ii. Pen Tool iii. Free Transform Tool
- Describe the steps to convert an object into a Symbol in Adobe Animate 2024.
- Differentiate between Fill and Stroke in a shape.



E. Competency-Based Questions

- Ayush has designed a company logo that he wants to use many times in his animation project. What should he do to avoid creating the same logo repeatedly? Explain your answer.
- Sneha wants to create a colourful background with a smooth blend of blue and green colours. Which feature of Adobe Animate should she use? Why?



Idea Lab

Imagine you have been asked to create a short animation to spread awareness about saving water. What objects and characters would you include? Draw a rough sketch of your idea and discuss it with your classmates.



Practical Exercises

- Find out about two famous animated movies and identify the software used to create their animations.
- Search the Internet and list any five additional tools available in Adobe Animate that were not discussed in this chapter.
- Create a colourful house using different shape tools and apply a gradient fill to at least one object.
- Design a simple logo, convert it into a Symbol, and insert it three times on the Stage.

Teacher's Note

- Demonstrate the basic tools and panels of Adobe Animate 2024 using a live classroom activity.
- Encourage students to create simple shapes and symbols to develop confidence before moving to animation.



Learning Goals

- Malware
- Types of Viruses
- How to prevent Virus
- Antiviruses
- Firewall



Let's Warm Up the Brain

What should you do when you fall sick? Tick () the correct picture.



A. ()



B. ()



C. ()



D. ()

Just as our body can become sick because of germs, a computer can also become “sick” because of harmful programs. These harmful programs can slow down the computer, damage files, or even steal important information. Just as medicines help us recover from illness, antivirus software and other security tools help protect our computers. In this chapter, we will learn about different types of malware and the ways to keep our computers safe.

Malware

A malware is a type of malicious program designed to damage a computer system or perform unwanted actions without the user's permission. The word ‘mal’ means bad, which makes it easy to remember that malware refers to harmful software. Malware may enter a computer through infected websites, e-mail attachments, unsafe downloads, or infected storage devices such as pen drives.

Once malware enters a computer, it can damage files, steal important information, slow down the system, or make the computer unusable. Different malware programs work in different ways, but all of them are harmful to the computer.

Some common types of malware are **worms**, **trojan horses**, **spyware**, **ransomware**, **rootkit**, and **computer viruses**.

Worms

A computer worm is a type of malware that has the ability to copy itself without any action from the user. It keeps creating copies of itself, which uses a large amount of the computer's memory and processing power.

When a computer is infected by a worm, it becomes slow, behaves unexpectedly, and may stop performing other tasks properly. A worm can also spread automatically from one computer to another through a network, making it one of the fastest spreading malware.



Trojan Horse

A Trojan Horse is a harmful program that hides inside software which appears to be genuine and useful. It tricks the user into installing it by pretending to be a safe application. Once the infected software is installed, the trojan becomes active and starts damaging the computer. It may steal information, delete files, or allow hackers to access the system secretly.

Spyware

Spyware is a type of malware that secretly watches the activities of a computer user. It collects information such as passwords, browsing history, and personal details without the user's knowledge. The collected information is then sent to hackers, who may misuse it. Since spyware works silently in the background, many users do not even know that their computer has been infected.

Ransomware

Ransomware is a dangerous type of malware that locks or encrypts the files stored on a computer. After locking the files, it demands money, called ransom, from the victim to restore access. If the ransom is not paid, the files may remain locked permanently. Ransomware can cause huge losses to individuals, schools, businesses, and organisations.

RootKit

A Rootkit is a type of malware that allows an attacker to gain administrator-level access to a computer. After entering the system, it hides itself so that it becomes difficult to detect. Even though it remains hidden, it continues to control the computer and gives special access to the attacker. Because of this, removing a rootkit is often very difficult.

Spyware

A computer virus is a piece of code or program developed to damage data or program files stored on a computer. It enters the computer without the user's permission and affects its normal working. A virus usually infects one computer at first. However, if the infected computer is connected to a network, the virus can spread to other computers. It reduces the processing speed of the system, damages files, and sometimes even corrupts the hard disk.

The word VIRUS stands for **Vital Information Resources Under Seize**.

Harms caused by Computer Viruses

Some common harmful effects of a computer virus are:

- It slows down the speed of the computer.
- It may delete or corrupt important files and folders.
- It can damage software and operating system files.
- It may display unwanted messages or advertisements.
- It can steal personal information such as passwords and banking details.
- It may cause the computer to crash or restart frequently.



Types of Computer Virus

Computer viruses are of different types depending on the way they infect a computer. Some viruses attack program files, while others infect the boot sector, documents, or e-mails. Each virus spreads in a different manner but all of them are harmful to the computer.

- Program File Virus
- Boot Sector Virus
- Macro Virus
- Email Virus

Program File Virus

A Program File Virus infects executable program files having extensions such as .exe and .com. When an infected program is opened, the virus becomes active and starts spreading to other program files stored on the computer. As more files become infected, the computer starts working slowly and some programs may stop working properly. This type of virus can also damage important software installed on the system.

Boot Sector Virus

A Boot Sector Virus attacks the boot sector of a storage device such as a hard disk or a pen drive. The boot sector contains important information that helps the computer start correctly. When the computer is switched on, the virus loads into the memory before the operating system starts. As a result, the computer may fail to boot properly or may show error messages during startup.

Macro Virus

A Macro Virus infects documents created using applications such as Microsoft Word and Microsoft Excel. It is written using macro programming commands and becomes active when the infected document is opened. This virus can spread from one document to another and may damage or modify the contents of the files. It can also spread through e-mail attachments or shared documents.

E-Mail Virus

An E-mail Virus spreads through infected e-mail messages and attachments. It usually arrives in the form of an attachment or a suspicious link sent by an unknown sender. When the attachment is opened or the link is clicked, the virus enters the computer and starts infecting files. It may also send copies of itself to the contacts stored in the user's e-mail account, allowing it to spread quickly.



Tech Fact

The Creeper program, created in 1971, is considered one of the earliest computer viruses. It displayed the message: **"I'm the Creeper, catch me if you can!"**. Later, another program called Reaper was developed to remove it, making it one of the first antivirus programs.



How does a Computer get infected with Virus?

Some common sources of computer viruses are:

- Downloading software from untrusted websites.
- Opening infected e-mail attachments.
- Clicking unknown or suspicious links.
- Using infected pen drives, CDs, or external hard disks.
- Visiting unsafe or malicious websites.
- Sharing infected files over a network.



How do you know Your PC has a Virus?

Some common symptoms of a virus infection are:

- The computer becomes unusually slow.
- Programs take a long time to open or stop responding.
- Frequent error messages appear on the screen.
- Files or folders disappear automatically.
- The computer restarts or crashes unexpectedly.
- Unwanted pop-up windows appear frequently.



How to prevent a Virus?

- Install a reliable antivirus program and keep it updated.
- Do not open e-mails from unknown senders.
- Scan pen drives and other storage devices before using them.
- Download software only from trusted websites.
- Avoid clicking suspicious links and advertisements.
- Keep the operating system and software updated regularly.
- Take regular backups of important files.



Antivirus

An antivirus is a software program specially designed to detect, prevent, and remove viruses and other types of malware from a computer. It regularly scans files, folders, programs, and storage devices to identify harmful software before it can damage the system. When a virus or malware is detected, the antivirus either removes it or places it in a secure area called quarantine, preventing it from spreading further.

New viruses are created every day, so antivirus companies regularly release updates to help their software recognize the latest threats. Therefore, it is important to keep the antivirus updated through the Internet. Most antivirus programs also provide real-time protection, which continuously monitors the computer and blocks suspicious files before they can infect the system. Using a reliable antivirus and scanning the computer regularly helps keep the system safe, secure, and running smoothly.



Some popular antivirus software are Norton, McAfee, Quick Heal, Kaspersky, Avast, AVG, Bitdefender, and Microsoft Defender.



Firewall

A firewall is a security system that protects a computer from unauthorised access through a network or the Internet. It acts like a security guard by checking all incoming and outgoing data and blocking suspicious connections. A firewall helps prevent hackers, viruses, and other malicious programs from entering the computer. Modern operating systems include built-in firewalls, but additional firewall software can also be installed for better protection.

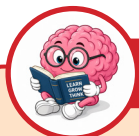


Activity

Let's Explore

Visit the official websites of any two antivirus companies. Find out:

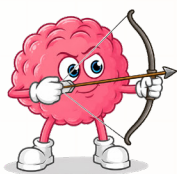
- Which operating systems they support.
- Any two features they provide.
- Whether they offer a free trial.



Quick Recap

- *Malware is a harmful software that damages a computer or steals information.*
- *Worms, Trojans, Spyware, Ransomware, Rootkits, and Viruses are common types of malware.*
- *A computer virus spreads by infecting files and programs.*
- *Antivirus software detects and removes harmful programs.*
- *A firewall protects a computer from unauthorised network access.*





TEST YOUR SKILLS



A. Choose the correct option

1. Which malware secretly collects a user's personal information?
a. Worm ☐ b. Spyware ☐
c. Firewall ☐ d. Antivirus ☐
2. Which of the following can protect a computer from malware?
a. Paint ☐ b. Calculator ☐
c. Antivirus ☐ d. Notepad ☐
3. Which malware locks files and asks for money to unlock them?
a. Trojan Horse ☐ b. Ransomware ☐
c. Worm ☐ d. Rootkit ☐
4. Which type of virus mainly infects Word and Excel documents?
a. Boot Sector Virus ☐ b. Macro Virus ☐
c. Program File Virus ☐ d. E-mail Virus ☐
5. Firewall _____ to protect a computer from viruses.
a. creating documents ☐ b. blocks unauthorised network ☐
c. deleting folders ☐ d. installing antiviruses ☐

B. Fill in the Blanks

1. A computer should always have an _____ installed to protect it from harmful programs.
2. A _____ secretly collects information from a user without permission.
3. Files locked by _____ can usually be accessed only after paying a ransom.
4. _____ checks incoming and outgoing network traffic to protect a computer.
5. A _____ can make copies of itself and spread to other computers.

C. State True or False

1. Antivirus software helps remove harmful programs. _____
2. A firewall increases the storage capacity of a computer. _____
3. Downloading files from trusted websites is a safe practice. _____
4. A worm needs the user to install it before it spreads. _____
5. Ransomware can lock important files on a computer. _____



D. Short Answer Type Questions

1. What is malware?
2. Name any four types of malware.
3. Why is antivirus software important?
4. What is the purpose of a firewall?

E. Long Answer Type Questions

1. Explain the difference between a computer virus and a worm.
2. Describe any four ways by which a computer can get infected with malware.
3. Explain the following:
 - i. Spyware
 - ii. RootKit

F. Competency-Based Questions

- a) Ayush downloaded a free game from an unknown website. After installing it, his computer became slow and unwanted advertisements started appearing on the screen. What could be the reason? What should he do now?
- b) Sneha received an e-mail from an unknown sender with an attachment named "Prize_List.pdf.exe". She is curious to open it. What advice would you give her? Explain your answer.
- c) Your school's computer lab is connected to the Internet. Suggest any five safety measures that should be followed to keep all the computers protected from malware.



Tech Practise

Create a table in MS Word with the following headings and fill it with suitable examples.

Malware	What it does	Example
Worm		
Trojan Horse		
Ransomware		
Spyware		
RootKit		





Word Search

Find the names of five cybersecurity terms hidden in the grid.

Malware () Firewall () Antivirus () Spyware () Trojan ()



Teacher's Note

- Explain the different types of malware using simple real-life examples and encourage students to discuss safe computing habits.
- Demonstrate how antivirus software scans a computer and explain the role of a firewall in protecting a system from online threats.



Introduction to HTML5 and CSS3

Learning Goals

- HTML
- Tags & Attributes
- HTML Structure
- Creating and Saving HTML Document
- Introducing CSS3
- Editing existing HTML Document



Let's Warm Up the Brain

Can you guess? Write your answers.

I open when you type a website name.

I connect one web page to another.

I am the language used to create web pages.

I am the unique address of a web page.

Computers connected through the Internet communicate with one another using networking protocols. Some computers act as servers, while others work as clients. A client sends a request to the server, and the server responds by providing the required information.

The information stored on a server is organised in the form of websites. A website is a collection of related web pages. A web page is a document that may contain text, pictures, audio, video and links to other web pages. Every web page has a unique address called the Uniform Resource Locator (URL), which helps users open the required page through a web browser. Web pages are created using HTML, the most widely used markup language for designing websites.

HTML

HTML stands for HyperText Markup Language. It is the standard markup language used to create the structure of web pages. HTML uses special tags to organise different elements such as headings, paragraphs, images, tables and hyperlinks. These tags provide instructions to the web browser on how the content should appear. HTML documents can be viewed on different operating systems using any modern web browser. The latest version of HTML is HTML5, which provides improved features for creating attractive and interactive web pages.

Hypertext

Hypertext is ordinary text that contains links to other documents or web pages. These links, called hyperlinks, allow users to move quickly from one web page to another by simply clicking on them.



Markup Language

A Markup Language is a language that uses special symbols called tags to describe the structure of a document. These tags instruct the web browser how the content should be displayed. The tags themselves are not visible on the web page but help in presenting the information correctly.

Key features of HTML5

Some of the important features of HTML5 are:

- It is a platform-independent language and can be used on different operating systems.
- It allows us to create tables, lists, forms, and multimedia elements on a web page.
- It supports hyperlinks, making it easy to connect one web page with another.
- It is not case-sensitive, can be written in uppercase, lowercase, or a combination of both.
- It is simple and easy to learn, making it suitable for beginners.
- It provides a flexible way to design well-structured and interactive web pages.

Types of HTML Editor

An HTML editor is a software application used to create and edit HTML documents. It helps users write HTML code and save it as a web page. HTML editors are mainly classified into two types: **WYSIWYG Editor** and **Text Editor**

WYSIWYG Editor

WYSIWYG stands for What You See Is What You Get. This type of editor allows users to design a web page visually without writing much HTML code. As the page is created, the editor displays how it will appear in a web browser. WYSIWYG editors provide various design tools, menus and graphical interfaces that make web page creation simple and convenient. Some popular WYSIWYG editors are Adobe Dreamweaver, Amaya, and Google Web Designer.

Text Editor

A Text Editor is a simple program used to create HTML documents by writing HTML code manually. It does not provide a graphical design interface, so the user should have a basic knowledge of HTML tags and commands. Common text editors used for HTML include Notepad, Notepad++, and WordPad. After writing the code, the file is saved with the .html extension and viewed in a web browser.

Tags and Attributes

HTML uses tags and attributes to create and format the contents of a web page. Tags define different elements such as headings, paragraphs, tables, images and links, while attributes provide additional information about these elements. Together, they help the web browser display the web page correctly.

Tags

An HTML tag is a special keyword enclosed within angle brackets (< >). Tags instruct the web browser how to display different elements on a web page. Most HTML tags are written in pairs, where the first tag is called the opening tag and the second tag is called the closing tag. The closing tag contains a forward slash (/) before the tag name.

Example: `<p>This is a paragraph.</p>`

Here, `<p>` is the opening tag and `</p>` is the closing tag.

Types of Tags

Container Tags: have both an opening tag and a closing tag. The content is written between these two tags. Most HTML tags are container tags.

Examples: `<html>...</html>`, `<head>...</head>`, `<p>...</p>`

Empty Tags: contains only an opening tag and do not require a closing tag. These tags do not enclose any content and are used to perform a specific task on a web page.

Examples: `<br>`, `<hr>`

Block Level Tags: occupy the entire width available on a web page and always start from a new line. These tags are mainly used to organise the structure of a web page.

Examples: `<h1>` to `<h6>`, `<p>`, `<div>`, `<hr>`

Text Level Tags: are used to format or highlight a part of the text. They do not begin on a new line and occupy only the required space.

Examples: `<b>`, `<i>`, `<u>`, `<sup>`, `<sub>`

Nesting of Tags

The process of placing one HTML tag inside another is called nesting of tags. Proper nesting helps maintain the correct structure of an HTML document. When tags are nested, the inner tag should be closed before closing the outer tag.

Attribute

Attributes provide additional information about an HTML element. They are written inside the opening tag and usually consist of a name and a value. Attributes modify the behaviour or appearance of an element and help in defining its properties.

The general syntax of an attribute is:

`<tagname attribute="value">Content</tagname>`

Example:

``

In this example, `src` and `alt` are attributes of the `<img>` tag. The `src` attribute specifies the location of the image, while the `alt` attribute displays alternate text if the image cannot be loaded.



Rules for Writing HTML5 Codes

While creating an HTML document, certain rules should be followed to ensure that the web page is displayed correctly in a web browser. These rules also make the HTML code simple, organised and easy to understand.

- Begin every HTML document with the `<!DOCTYPE html>` declaration.
- Enclose all HTML code within the `<html>` and `</html>` tags.
- Write the information about the web page inside the `<head>` section and the visible content inside the `<body>` section.
- Close all paired tags in the correct order.
- Use lowercase letters for writing HTML5 tags and attributes.
- Save the file with the `.html` or `.htm` extension.

HTML5 Document Structure

Every HTML5 document follows a fixed structure. It consists of different tags that define various sections of a web page. The basic structure includes the `<!DOCTYPE html>`, `<html>`, `<head>`, `<title>`, and `<body>` tags. Each tag performs a specific function and helps the browser interpret the document correctly.

The basic structure of an HTML5 document is shown below:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>My First Web Page</title>
5 </head>
6 <body>
7   Welcome to HTML5!
8 </body>
9 </html>
```

<!DOCTYPE html> Tag

The `<!DOCTYPE html>` declaration appears at the beginning of an HTML document. It informs the web browser that the document is written in HTML5. This declaration helps the browser display the web page according to the HTML5 standard.

<HTML> Tag

The `<html>` tag is the root element of an HTML document. It contains all the other HTML elements and indicates the beginning of the web page. Every HTML document starts with the `<html>` tag and ends with the `</html>` tag.

<HEAD> Tag

The <head> tag contains information about the web page that is not displayed on the browser window. It includes the page title, character encoding, links to style sheets, scripts and other important information required by the browser.

<TITLE> Tag

The <title> tag is placed inside the <head> section. It specifies the title of the web page, which appears on the title bar or browser tab. A meaningful title also helps users identify the web page easily.

<BODY> Tag

The <body> tag contains all the content that is displayed on the web page. Text, headings, images, tables, lists, hyperlinks and other multimedia elements are placed inside the body section. Everything written between the <body> and </body> tags becomes visible in the browser window.



Activity

Challenge:

Write a short paragraph about your favourite hobby using HTML tags. Use at least one block-level tag, two text-level tags, and one empty tag. Save the file and view the output in a web browser.

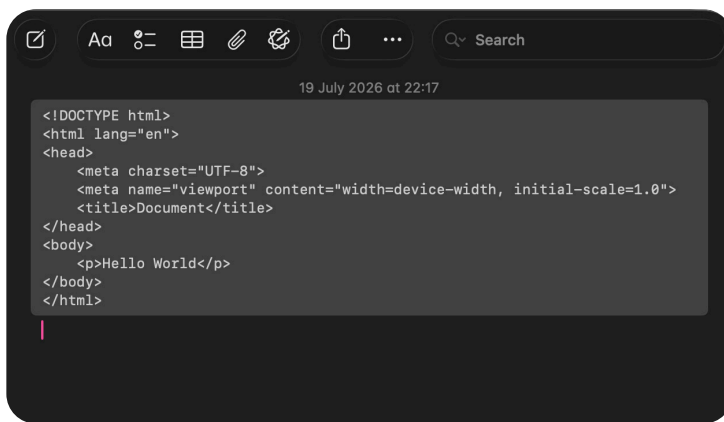
Creating and Saving an HTML Document

An HTML document can be created using any text editor, such as Notepad or Notepad++. After writing the HTML code, the file should be saved with the .html or .htm extension. Saving the file with the correct extension allows a web browser to recognise it as an HTML document and display it properly.

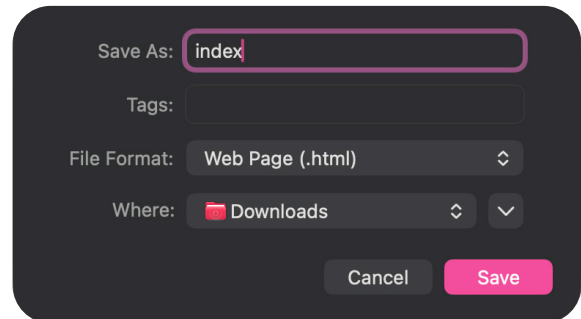
Follow these steps to create and save an HTML document:

- Open Notepad or any text editor.
- Type the required HTML code.
- Click the File menu and select Save As.
- Enter a suitable file name followed by the .html extension. For example, MyPage.html.
- Select All Files from the Save as type list.
- Click the Save button.





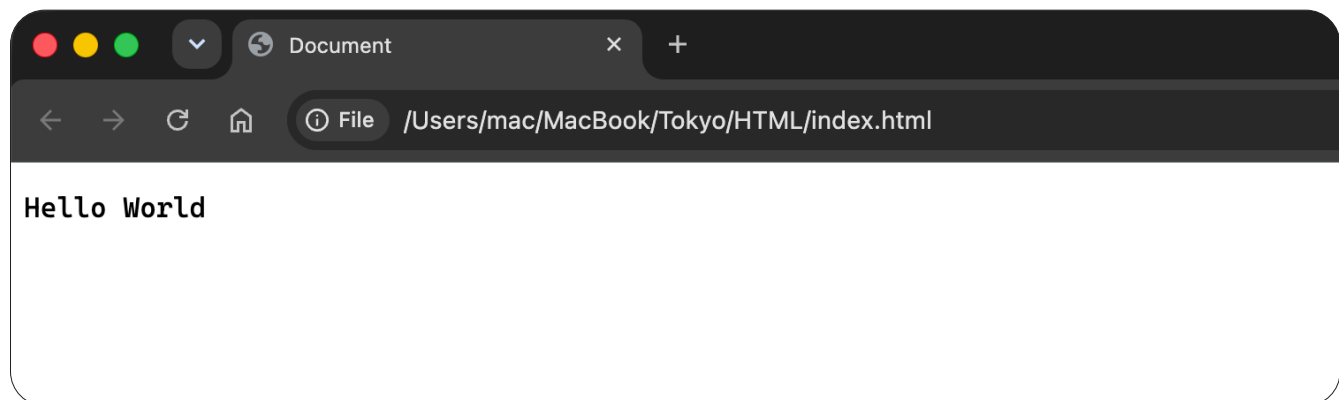
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <p>Hello World</p>
</body>
</html>
```



Viewing a Web Page

After creating and saving an HTML document, it can be viewed in a web browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari. The browser reads the HTML code and displays the web page in a formatted manner.

To open a web page, locate the saved HTML file and double-click it. The default web browser will open the file and display its contents. If any changes are made to the HTML code, save the file again and refresh the browser window to view the updated page.



Introducing CSS3

CSS stands for Cascading Style Sheets. It is a style sheet language used to control the appearance and layout of HTML web pages. CSS helps make web pages more attractive by changing the colours, fonts, backgrounds, borders, spacing and alignment of different elements.

The latest version of CSS is CSS3. It provides many advanced features such as gradients, shadows, animations and rounded corners, making it easier to design modern and responsive web pages. CSS works together with HTML, where HTML defines the structure of a web page and CSS controls its presentation.

Using CSS also saves time because the same style can be applied to multiple web pages, giving a website a consistent and professional appearance.



Using CSS Style

CSS styles can be applied to an HTML document in different ways. Depending on the requirement, a style can be written directly inside an HTML element, within the HTML document itself, or in a separate CSS file. The three commonly used methods of applying CSS are:

- Inline Style Sheet
- Internal Style Sheet
- External Style Sheet

Inline Style CSS

An Inline Style Sheet is used to apply CSS to a single HTML element. The style is written directly inside the opening tag of the element using the style attribute. It affects only that particular element and does not influence any other part of the web page. Inline styles are useful when a specific element needs a unique appearance.



```
1 <p style="color:blue; font-size:20px;">
2 Welcome to HTML5
3 </p>
```

Internal Style CSS

After creating and saving an HTML document, it can be viewed in a web browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari. The browser reads the HTML code and displays the web page in a formatted manner.

To open a web page, locate the saved HTML file and double-click it. The default web browser will open the file and display its contents. If any changes are made to the HTML code, save the file again and refresh the browser window to view the updated page.



```
1 <head>
2 <style>
3 h1{
4     color:green;
5 }
6 p{
7     font-size:18px;
8 }
9 </style>
10 </head>
```



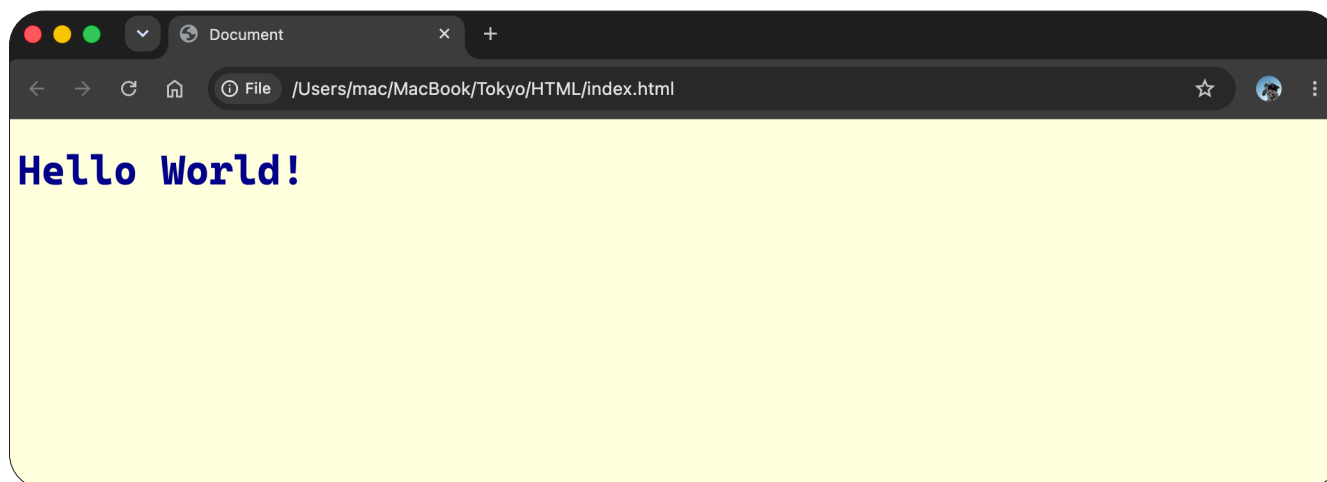
External Style CSS

After creating and saving an HTML document, it can be viewed in a web browser such as Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari. The browser reads the HTML code and displays the web page in a formatted manner.

To open a web page, locate the saved HTML file and double-click it. The default web browser will open the file and display its contents. If any changes are made to the HTML code, save the file again and refresh the browser window to view the updated page.

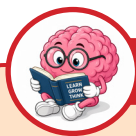
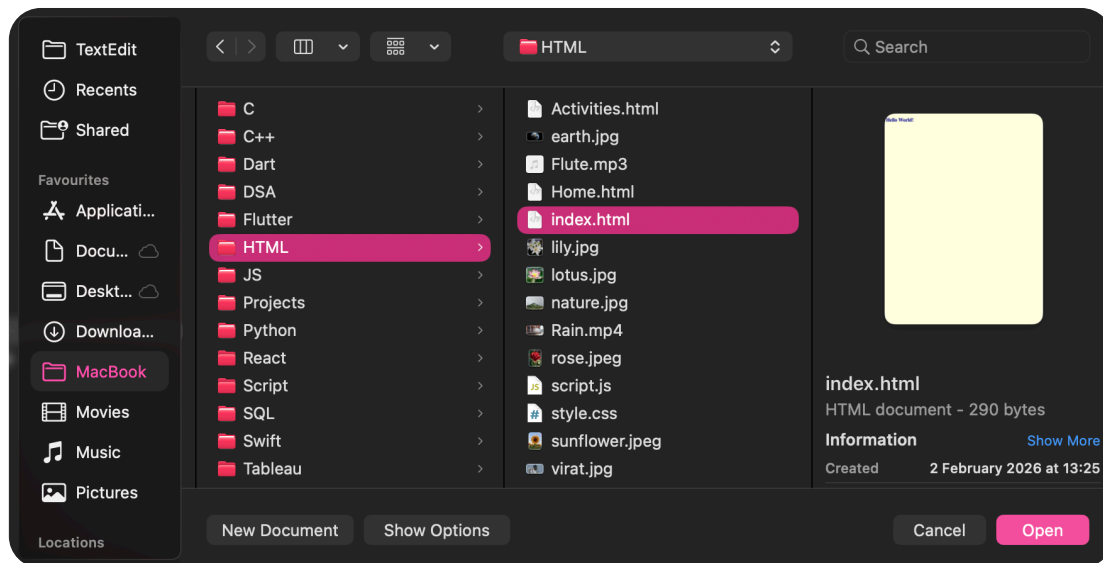
```
1 <!-- HTML File -->
2
3 <!DOCTYPE html>
4 <html lang="en">
5 <head>
6     <meta charset="UTF-8">
7     <meta name="viewport" content="width=device-width, initial-scale=1.0">
8     <title>Document</title>
9 </head>
10 <body>
11     <h1>Hello World!</h1>
12 </body>
13 </html>
```

```
1 <!-- CSS File -->
2
3 body{ background-color: lightyellow; }
4 h1{ color: darkblue; }
```



Editing an Existing HTML Document

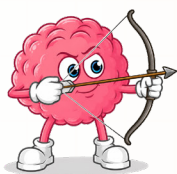
An HTML document can be edited at any time using a text editor. Open the existing HTML file, make the required changes to the code, and save the file again. After saving, refresh the web page in the browser to view the updated content. Editing an existing document is useful for correcting errors, updating information, or improving the design of a web page without creating a new file.



Quick Recap

- *HTML5 is the standard markup language used to create web pages.*
- *Hypertext links one web page to another through hyperlinks.*
- *Tags are the basic building blocks of an HTML document.*
- *Attributes provide additional information about HTML tags.*
- *An HTML document is mainly divided into the Head and Body sections.*
- *CSS3 is used to improve the appearance of web pages.*
- *CSS styles can be applied using Inline, Internal, and External style sheets.*
- *HTML files are saved with the .html extension and viewed in a web browser.*





TEST YOUR SKILLS



A. Choose the correct option

- Which tag contains the visible content of a web page?
a. <head> () b. <body> ()
c. <title> () d. <html> ()
- HTML files are usually saved with which extension?
a. .doc () b. .txt ()
c. .html () d. .css ()
- Which CSS method stores styles in a separate file?
a. Inline Style Sheet () b. Internal Style Sheet ()
c. External Style Sheet () d. Embedded Style ()
- Which tag is used to connect an external CSS file with an HTML document?
a. <style> () b. <css> ()
c. <link> () d. <script> ()
- Which software is mainly used to view an HTML page?
a. Web Browser () b. Calculator ()
c. Paint () d. Media Player ()

B. Fill in the Blanks

- The latest version of HTML is _____.
- A unique address of a web page is called a _____.
- _____ is used to improve the appearance of a web page.
- Additional information about an HTML tag is provided by an _____.
- An HTML page is displayed using a web _____.

C. State True or False

- CSS is used to create the structure of a web page. _____
- HTML tags are enclosed within angle brackets. _____
- The <title> tag is placed inside the <head> section. _____
- An external style sheet is saved with the .css extension. _____
- Hyperlinks connect one web page to another. _____



D. Short Answer Type Questions

1. What is Hypertext?
2. What is the purpose of HTML?
3. What is an HTML attribute?
4. Name any two HTML editors.
5. What is the use of the <title> tag?

E. Long Answer Type Questions

1. Explain the basic structure of an HTML document with the help of a diagram or code.
2. Differentiate between Inline, Internal, and External Style Sheets.
3. Explain the steps involved in creating and saving an HTML document.

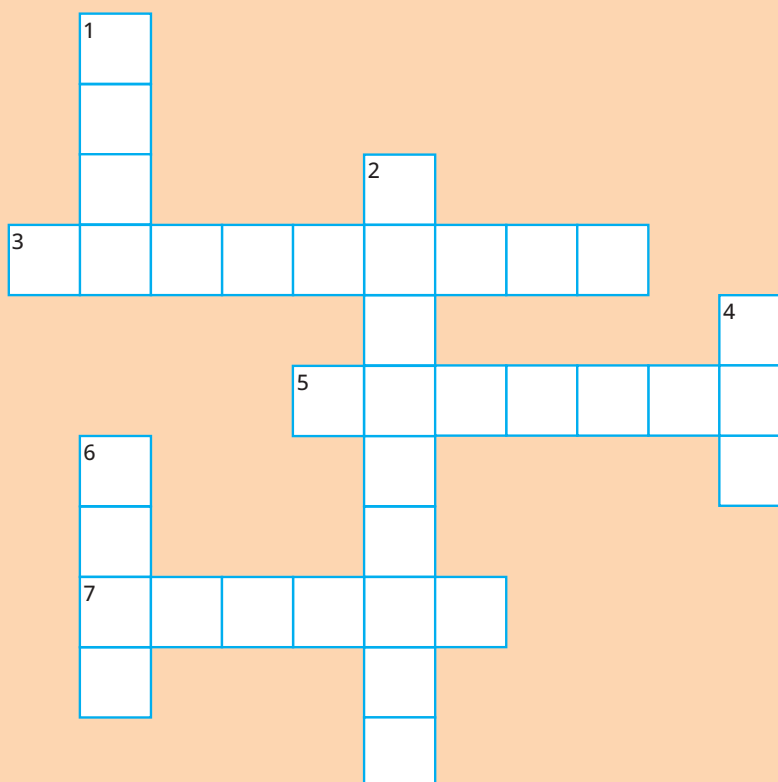
F. Competency-Based Questions

- a. Rahul has created a web page but wants the same design to be used on all the pages of his website. Which type of CSS should he use? Give a reason for your answer.
- b. Meera accidentally saved her HTML file with the .txt extension. The file is not opening as a web page in the browser. What should she do to solve the problem?



Practical Exercises

Let's Solve:



Across:

3. Text that links one web page to another.
5. Software used to view a web page.
7. A language that uses tags to describe the structure of a document.

Down:

1. The visible part of an HTML document.
2. Additional information provided to an HTML tag.
3. A unique address of a web page.
4. Language used to create the structure of web pages.



SDG Activity

Design an HTML and CSS web page on the topic “Digital Learning for Everyone.” Include a heading, a short paragraph explaining the importance of online learning, and use suitable colours to make the page attractive.

Teacher's Note

- *Demonstrate the use of different HTML tags and CSS styles by creating a sample web page in class.*
- *Encourage students to experiment with different colours, fonts, and layouts while designing their own web pages.*



Learning Goals

- Algorithms
- Characteristics of good Algorithm
- Uses of Algorithm
- Writing an Algorithm
- Defining Flowcharts
- Solve Problems using Algorithms and Flowcharts



Let's Warm Up the Brain

Look at the code and decode the message.

Code:

A-1, B-2, C-3, D-4, E-5, F-6, G-7, H-8, I-9, J-10, K-11, L-12, M-13, N-14, O-15, P-16, Q-17, R-18, S-19, T-20, U-21, V-22, W-23, X-24, Y-25, Z-26, Space-27

Message:

3	15	4	5	27	19	13	1	18	20	27	20	8	9	14	11	9	14	7

In our daily life, we perform many activities by following a sequence of steps. For example, preparing a meal, solving a mathematics problem, or planting a tree requires us to carry out one step after another in the correct order. In the same way, a computer also needs clear and well-organised instructions to perform any task. These instructions help the computer understand what to do and how to do it.

Before writing a computer program, a programmer first plans the solution to the problem. This planning is done using algorithms and flowcharts. An algorithm describes the steps needed to solve a problem, while a flowchart represents these steps using standard symbols. Together, they make problem-solving easier, reduce mistakes, and help in developing accurate computer programs.

Algorithm

A program is a collection of instructions written in a programming language that tells a computer how to perform a specific task. Before writing a program, it is important to understand the problem carefully and decide the correct sequence of steps required to solve it. These steps are organised in the form of an algorithm.

An algorithm is a set of simple, clear, and sequential steps used to solve a problem or complete a task. It is written in easy-to-understand language so that anyone can follow the instructions without confusion. Every algorithm has a definite starting point and an ending point, and each step is performed in the given order.

In computer terms, an algorithm is a sequence of logical steps or computer operations that are followed to solve a given problem and produce the desired result.



Characteristics of a Good Algorithm

- **Precision:** Every step of the algorithm should be clear, accurate, and precisely defined so that there is no confusion while performing the task.
- **Uniqueness:** Each step should produce a definite result and should depend only on the outcome of the previous step.
- **Finiteness:** A good algorithm must end after executing a finite number of steps. It should not continue endlessly.
- **Input:** The algorithm should accept one or more well-defined inputs required to solve the problem.
- **Output:** It should produce at least one well-defined output that provides the solution to the problem.
- **Effectiveness:** Every instruction in the algorithm should be simple, practical, and possible to execute within a reasonable amount of time and memory.

Uses of an Algorithm

Algorithms are widely used in computer science as well as in everyday problem-solving. Some common uses of an algorithm are:

- Performing data processing.
- Carrying out computer and mathematical calculations.
- Manipulating and organising data.
- Searching for a particular piece of information.
- Solving logical and real-life problems in a systematic manner.
- Developing computer programs and software applications.

Writing an Algorithm

Writing an algorithm requires careful thinking and planning. Before preparing an algorithm, the problem should be analysed properly so that the correct sequence of steps can be arranged. The instructions should be written in simple language, and each step should clearly describe the action to be performed.

A good algorithm begins with the Start step and ends with the Stop step. Between these two steps, the required inputs are taken, necessary operations are performed, decisions are made if required, and finally the desired output is displayed. A properly written algorithm makes programming easier and helps in finding errors before writing the actual program.

A. Algorithm to Find the Product, Sum, and Difference of Two Numbers

The following algorithm shows how to calculate the product, sum, and difference of two numbers. First, the two numbers are taken as input and stored in variables. The required mathematical operations are then performed one by one, and the results are displayed.



Algorithm

Step 1: Start.

Step 2: Read two numbers and store them in variables A and B.

Step 3: Multiply A and B.

Step 4: Display the product.

Step 5: Add A and B.

Step 6: Display the sum.

Step 7: Subtract B from A.

Step 8: Display the difference.

Step 9: Stop.

B. Algorithm to Find the Smaller of Two Numbers

This algorithm is used to compare two numbers and determine which one is smaller. The two numbers are first taken as input and then compared with each other. Based on the comparison, the smaller number is displayed.

Algorithm

Step 1: Start.

Step 2: Read two numbers and store them in variables A and B.

Step 3: Compare A and B. If A is less than B, display "A is smaller." Then go to Step 5.

Step 4: If B is less than A, display "B is smaller."

Step 5: Stop.

Defining Flowcharts

A flowchart is a graphical representation of an algorithm. It uses different standard symbols connected by arrows to show the sequence of steps required to solve a problem. Each symbol represents a particular action, such as taking input, performing a process, making a decision, or displaying the output.

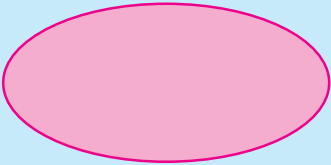
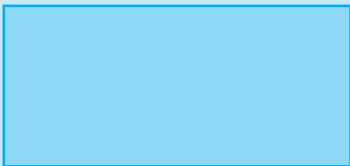
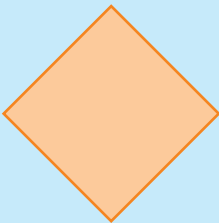
Flowcharts make it easier to understand the logic of a program before writing the actual code. They help programmers analyse a problem, organise the sequence of operations, and identify possible errors at an early stage. Since flowcharts present information visually, they are easy to read and explain, even for complex problems.

Flowcharts are widely used in planning, designing, documenting, and managing programs and processes in various fields. Once a flowchart is prepared, it becomes easier to convert the solution into a computer program using any programming language.



Symbols used in Flowcharts

Different standard symbols used while drawing a flowchart are:

Symbol	Name	Description
	(Start/Stop)	It is used to indicate the beginning and the end of a flowchart. It usually contains the words Start or Stop.
	Process	It represents a processing or action step where calculations or operations are performed. This is the most commonly used flowchart symbol.
	Decision	It represents a condition or question that requires a decision. It usually has two possible paths such as Yes/No or True/False.
	Input/Output	It is used to show data entering the system as input or information leaving the system as output.
	Flow Line (Arrow)	It shows the direction in which the flow of the process moves from one step to another.
	Connector (Inspection)	It is used to connect different parts of a flowchart. It helps continue the flow when the chart is too large or when a jump from one point to another is required.

Rules to draw a Flowchart

A flowchart should follow certain rules to make it clear and easy to understand.

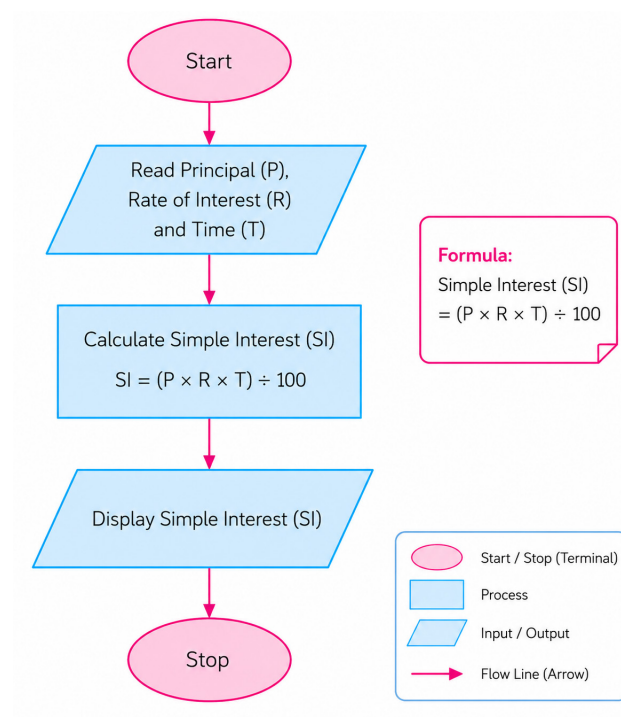
- Draw the flowchart neatly and clearly.
- Maintain the direction of flow from top to bottom or left to right.
- Only one flow line should come out from a process symbol.
- A decision symbol should have one incoming flow line and two outgoing flow lines representing different outcomes.
- Use standard flowchart symbols correctly.
- Ensure that every flowchart has a proper Start and Stop point.
- Keep the sequence of steps logical and easy to follow.

Advantages of Flowchart

Flowcharts are useful because they represent the solution to a problem in a simple and visual manner. Some important advantages of flowcharts are:

- They provide a better understanding of a problem.
- They help programmers analyse a problem in detail.
- They make the sequence of steps easy to understand.
- They simplify the process of writing computer programs.
- They make programs easier to explain, modify, and maintain.

Example: Flowchart to calculate Simple Interest



Solving problems using Algorithm and Flowchart

Example 1: Algorithm to Find the Factorial of a Number

The factorial of a number is the product of all positive integers from 1 to that number. It is represented by the symbol (!). For example,

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

The following algorithm calculates the factorial of a given number.

Algorithm

Step 1: Start.

Step 2: Read the number N.

Step 3: Set $i = 1$ and $\text{Fact} = 1$.

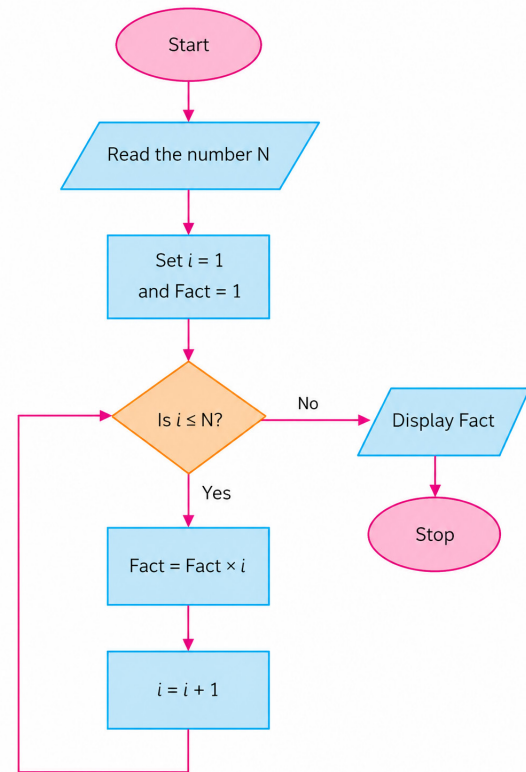
Step 4: Check whether $i \leq N$. If true, go to Step 5; otherwise, go to Step 7.

Step 5: Multiply Fact by i and store the result in Fact.

Step 6: Increase i by 1 and go back to Step 4.

Step 7: Display the value of Fact.

Step 8: Stop.



Example 2: Algorithm to Check Whether a Number is Even or Odd

A number is even if it is exactly divisible by 2. Otherwise, it is an odd number. This algorithm checks the remainder after dividing the number by 2 and displays the appropriate result.

Algorithm

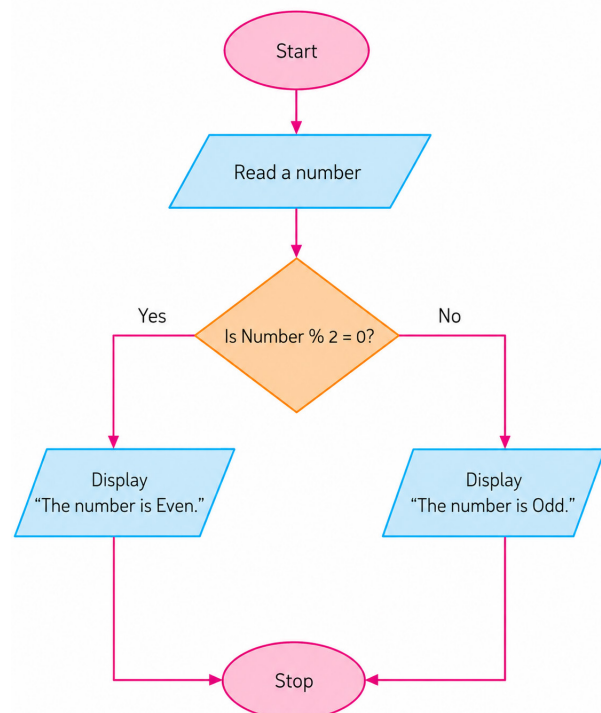
Step 1: Start.

Step 2: Read the number N.

Step 3: Check whether $\text{Number} \% 2 = 0$.

- If Yes, display "The number is Even."
- Otherwise, display "The number is Odd."

Step 4: Stop.

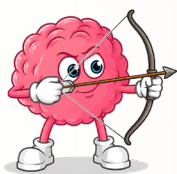




Quick Recap

- A program is a set of instructions that tells a computer how to perform a specific task.
- An algorithm is a step-by-step procedure used to solve a problem in a logical and systematic manner.
- Algorithms are used in problem-solving, programming, calculations, data processing, and decision-making.
- A flowchart is a graphical representation of an algorithm using standard symbols.
- Flowcharts make it easier to understand, analyse, and communicate the logic of a program.
- Algorithms and flowcharts together make program development easier, faster, and more organised.





TEST YOUR SKILLS



A. Choose the correct option

- A step-by-step procedure used to solve a problem is called a:
a. Program () b. Algorithm ()
c. Compiler () d. Software ()
- Which flowchart symbol is used to represent a decision?
a. Rectangle () b. Oval ()
c. Diamond () d. Arrow ()
- Which symbol is used to show input and output operations?
a. Oval () b. Parallelogram ()
c. Rectangle () d. Diamond ()
- A good algorithm must always have:
a. Infinite steps () b. Definite beginning & ending ()
c. Only one output () d. No output ()
- Which of the following is a characteristic of a good algorithm?
a. Confusing () b. Endless ()
c. Precise () d. Random ()

B. Fill in the Blanks

- A _____ is a graphical representation of an algorithm.
- The _____ symbol is used to perform calculations in a flowchart.
- Every algorithm begins with Start and ends with _____.
- The _____ symbol is used for taking input and displaying output.
- A good algorithm should produce the correct _____.

C. State True or False

- A flowchart uses standard symbols to represent different operations. _____
- An algorithm may continue forever without stopping. _____
- The diamond symbol represents a decision in a flowchart. _____
- A flowchart makes a program difficult to understand. _____
- Algorithms help programmers plan the solution before coding. _____



D. Short Answer Questions

1. What is an algorithm?
2. Name any four characteristics of a good algorithm.
3. Why are algorithms useful before writing a program?

E. Long Answer Questions

1. Explain the characteristics of a good algorithm.
2. Describe the different symbols used in a flowchart with their functions.
3. Write the rules that should be followed while drawing a flowchart.
4. Explain the advantages of using flowcharts.
5. How do algorithms and flowcharts help in problem-solving?



Activity

Create a flowchart to display whether a student's marks are Pass or Fail. If the marks are 40 or above, display "Pass"; otherwise, display "Fail."

A large, empty rectangular box with rounded corners, intended for drawing a flowchart.



Explore

Search the Internet or visit your school library to learn about different programming languages used for developing computer programs. Prepare a short report on any three programming languages, mentioning their year of development, creator, and one common use of each language. Share your findings with your classmates.



Activity

Many activities in our daily life follow a sequence of steps. Choose any three of the following activities and write an algorithm for it:

- Preparing a sandwich
- Watering plants
- Packing a school bag
- Brushing your teeth
- Making a glass of lemonade

Then, draw a flowchart for the same activity using the correct flowchart symbols.

Teacher's Note

Explain the importance of algorithms, flowcharts, and binary representation in solving problems. Encourage students to write simple IF-THEN-ELSE conditions using real-life situations before introducing programming languages.



Introduction to Programming

Learning Goals

- Computer Languages
- Language Translation
- Python
- Features of Python
- Python Installation
- Programming in Python
- Python Programs



Let's Warm Up the Brain

State whether the following statements are True or False.

- Computers can perform tasks only when they are given instructions.

- Different programming languages are used to create computer programs. _____
- A computer can understand human language directly without any programming language. _____

A program is a set of instructions written to make a computer perform a particular task. The process of writing these instructions is called programming. Programming helps us create useful software such as games, calculators, drawing applications, websites, and many other computer programs. A person who writes computer programs is called a programmer.

Different programming languages are used to write programs for different purposes. Some languages are designed to create websites, while others are used for making games, mobile apps, or scientific applications. In this chapter, we will learn about computer languages, language translators, Python programming, variables, operators, and many other basic concepts of programming.

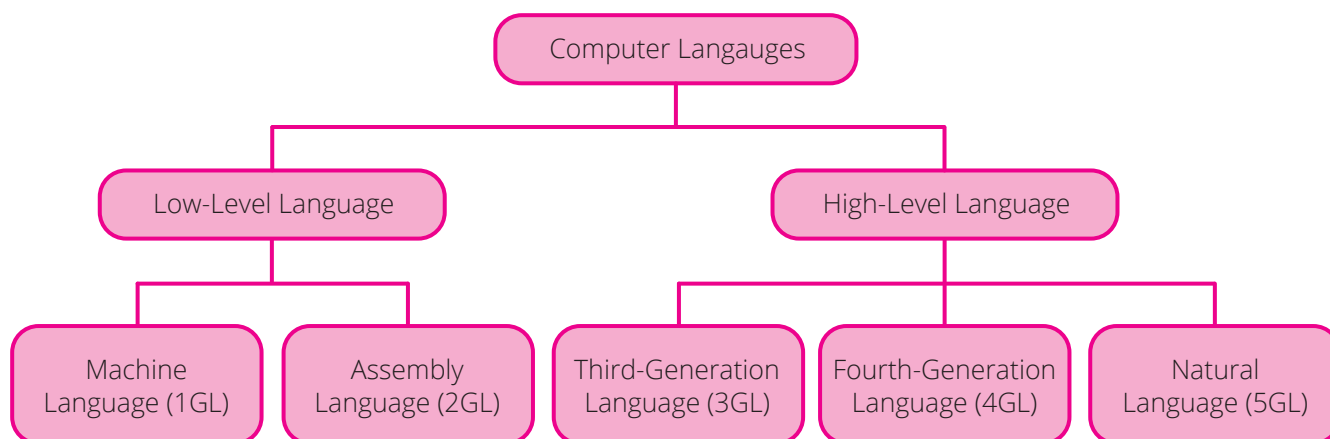
Conditional Statements

People communicate with one another using different languages such as English, Hindi, French, or Spanish. In the same way, we need a special language to communicate with a computer. These special languages are called computer languages or programming languages.

A computer language is a language used to write instructions that tell the computer what to do. These instructions are arranged in a logical order so that the computer can perform a specific task correctly. Without programming languages, it would be impossible to create software or applications. Programming languages contain words, symbols, numbers, and rules that help programmers write instructions. Every programming language has its own syntax (rules for writing programs), just as every spoken language has its own grammar.

Modern computers support many advanced programming languages. Computer languages are broadly divided into two main categories:





Low-Level Language (LLL)

A Low-Level Language (LLL) is a programming language that is very close to the language understood by a computer. It is called machine-dependent because a program written for one type of computer may not work on another type without changes.

Low-level languages provide faster execution because they are closely related to the computer's hardware. However, they are difficult for humans to read, write, and understand. Therefore, they are generally used by expert programmers.

Machine Language

Machine Language is the first generation programming language and is the only language that a computer understands directly. It consists entirely of binary digits (0 and 1). These binary digits are called bits.

Since machine language uses only 0s and 1s, writing programs in it is very difficult. Even a small mistake can cause the program to work incorrectly. It is also time-consuming and hard to remember long sequences of binary numbers.

Example: 10110011 00011100 11101010

Assembly Language

Assembly Language is the second generation programming language. Instead of writing long binary numbers, programmers use short words called mnemonics to represent instructions. These mnemonics make programs easier to read and write than machine language. However, assembly language is still machine-dependent, which means it is written for a particular type of computer.

Since computers cannot understand assembly language directly, it must first be converted into machine language using a special translator called an Assembler.

Example:

- **ADD:** Add two values
- **MUL:** Multiply values
- **MOV:** Move Data
- **SUB:** Subtract values
- **DIV:** Divide values



High-Level Language (HLL)

A High-Level Language (HLL) is a programming language that is closer to human language than machine language. It uses English-like words and simple rules, making it easier to learn, write, and understand. Unlike low-level languages, high-level languages are generally machine-independent, which means the same program can run on different computers after translation. High-level languages reduce the effort required to write programs and make it easier to find and correct mistakes. Because of these advantages, they are widely used for developing software today.

Third-Generation Language

Third Generation Languages use English-like statements and mathematical operators. In these languages, programmers write step-by-step instructions to solve a problem.

Example: C, C++, Java, BASIC, Pascal

Fourth-Generation Language

Fourth Generation Languages are designed to make programming easier by allowing programmers to describe what they want instead of explaining every step. These languages reduce programming time and require fewer lines of code.

Example: SQL, Matlab, SAS

Fifth- Generation (Natural) Language

Fifth Generation Languages are mainly used in Artificial Intelligence (AI) and expert systems. They are designed to solve complex problems using advanced technologies. These languages focus on intelligent decision-making and are commonly used in research and modern AI applications.

Example: Prolog, Mercury

Language Translator

A computer understands only machine language, which is made up of binary digits (0s and 1s). However, programmers usually write programs in high-level or assembly languages because they are much easier to read and write. Since a computer cannot understand these languages directly, the program must first be converted into machine language before it can be executed. This conversion is done by a special software called a language translator.

A language translator is a system software that converts a program written in a programming language into machine language so that the computer can understand and execute it. Without a language translator, the computer would not be able to run programs written by programmers.

There are three main types of language translators: **Assembler, Compiler, Interpreter**



Assembler

An Assembler is a language translator that converts a program written in assembly language into machine language. Assembly language uses short words called mnemonics, such as ADD, SUB, and MOV, instead of long binary numbers. The assembler changes these mnemonics into machine code that the computer can understand. Assemblers are mainly used for programs that need direct communication with computer hardware.

Compiler

A Compiler is a language translator that converts the entire high-level language program into machine language at one time. Before the program is executed, the compiler checks the whole program for errors. If there are mistakes, it displays a list of errors so that the programmer can correct them. After all the errors are removed, the compiler generates machine code, and the program is ready to run.

Interpreter

An Interpreter is a language translator that converts and executes a program one statement at a time. It reads the first line of the program, translates it into machine language, executes it, and then moves to the next line. If an error is found, the interpreter stops immediately and displays the error. The remaining lines are executed only after the mistake has been corrected.

Python

Python is one of the most popular and easy-to-learn programming languages in the world. It was created by the Dutch programmer Guido van Rossum and was first released in 1991.

Python is a high-level, general-purpose programming language. It uses simple English-like commands, making it easy for beginners to understand. Because of its simple syntax and powerful features, Python is widely used by students, teachers, software developers, scientists, and engineers.

Today, Python is used in many fields, including education, web development, artificial intelligence, robotics, automation, and data analysis.

Features of Python

- **Easy to learn:** Python uses simple English-like statements that are easy to read and understand. Beginners can quickly learn how to write programs without memorising complicated rules.
- **Easy to Code:** Python programs usually require fewer lines of code than many other programming languages. This makes programs shorter, clearer, and easier to understand.
- **Free & Open-Source:** Python is available free of cost. Anyone can download, use, and improve it according to the rules of its open-source licence.
- **Cross-Platform:** Python programs can run on different operating systems such as Windows, macOS, and Linux with very little or no modification.



- **Object-Oriented:** Python supports Object-Oriented Programming (OOP), which helps programmers organise programs into reusable objects and classes. This makes large programs easier to manage.
- **Interpreted Language:** Python uses an interpreter, which executes programs one statement at a time. This allows programmers to identify and correct errors easily during program execution.
- **Large Library Support:** Python provides thousands of ready-made libraries that help programmers perform tasks such as drawing graphics, working with files, creating games, performing calculations, and developing AI applications without writing everything from scratch.

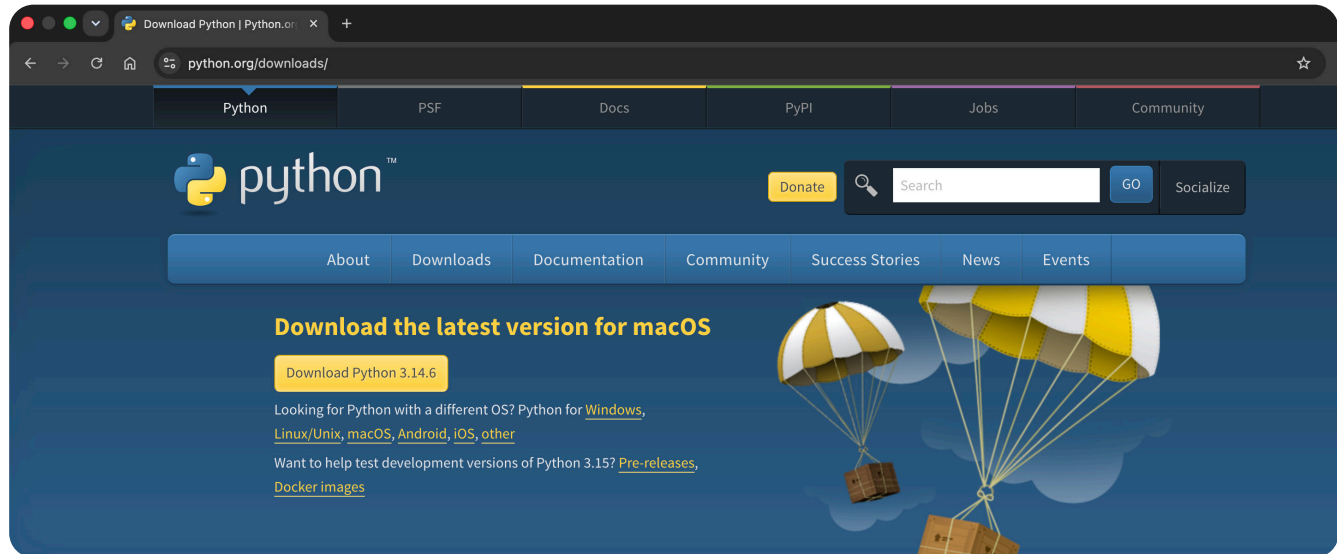
Installing Python

Before writing Python programs, we need to install the Python software on our computer. The official Python software can be downloaded free of cost from the Python website.

Follow these steps to install Python:

Step 1: Open a web browser and visit the official Python website.

Step 2: Open the Downloads section and download the latest stable version of Python suitable for your operating system.



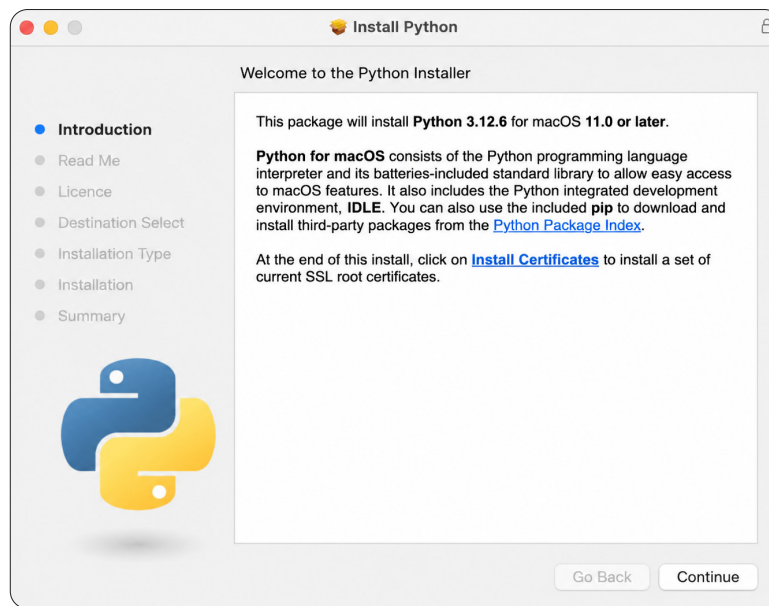
Step 3: After the download is complete, open the downloaded installation file.

Step 4: The Python Setup window will appear. Select the option Add Python to PATH before starting the installation.

Step 5: Click the Install Now button and wait while Python is installed on your computer.

Step 6: After the installation is complete, click Close. Python is now ready to use.





Programming in Python

After installing Python on a computer, we can begin writing and executing Python programs. Python provides a simple and friendly environment that helps beginners learn programming easily. It offers two different ways to write and run programs. These are called Interactive Mode and Script Mode. Both modes are useful, but they are used for different purposes.

Working in Interactive Mode

Interactive Mode allows us to type one command at a time. As soon as we press the Enter key, Python immediately executes the command and displays the result. This mode is useful for performing quick calculations, testing small pieces of code, and learning Python commands.

Follow these steps to open Python in Interactive Mode:

Step 1: Open the Start menu (or Applications folder on macOS/Linux).

Step 2: Search for Python IDLE.

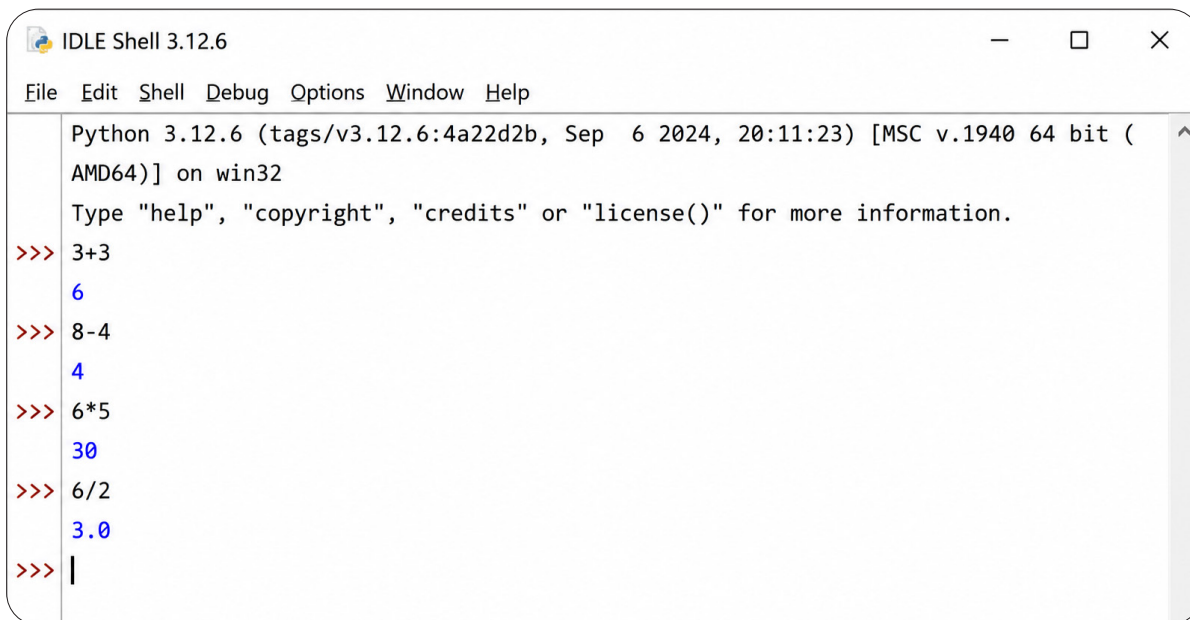
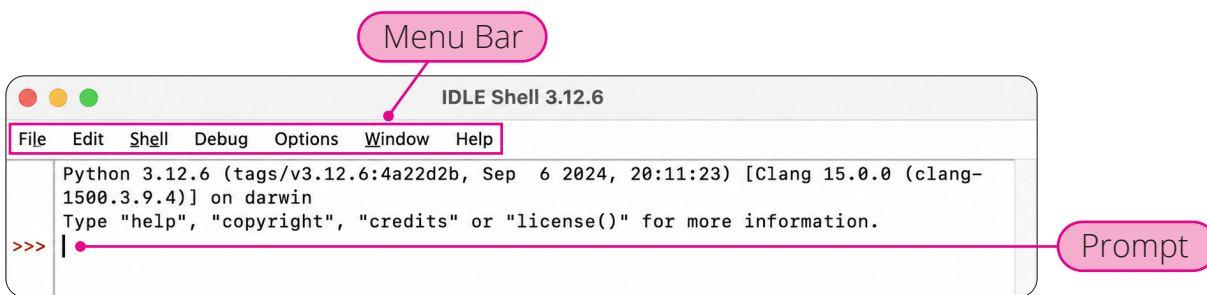
Step 3: Click IDLE (Python).

Step 4: The Python Shell window opens with the prompt `>>>`, indicating that Python is ready to accept commands.

Components of Python Window

Menu Bar: The Menu Bar appears at the top of the window. It contains different menus such as File, Edit, Shell, Debug, Options, Window, and Help. These menus allow us to create new files, save programs, run programs, and change settings.

Prompt: The symbol `>>>` is called the Python Prompt. It shows that Python is waiting for a command. Whatever we type after the prompt is executed when we press the Enter key.



Working in Strict Mode

While Interactive Mode is useful for trying small commands, it is not suitable for writing long programs because the commands are not saved. For creating complete programs, Python provides Script Mode. In Script Mode, we write all the instructions in a file, save the file, and run it whenever required. The program remains stored on the computer and can be edited or executed again later.

Most Python applications are created using Script Mode.

Creating a New File

To write a Python program, we first need to create a new file.

Follow these steps:

Step 1: Open Python IDLE.

Step 2: Click the File menu.

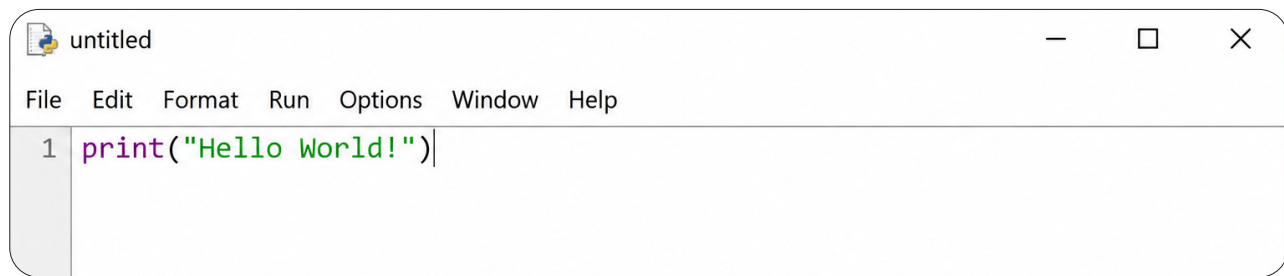
Step 3: Select New File.

Step 4: A blank Editor window appears where you can type your Python program.



Writing a Python Program

Now we can type our program in the Editor window.



Saving Python Program

After writing a program, it should be saved so that it can be used again later.

Follow these steps:

Step 1: Click the File menu.

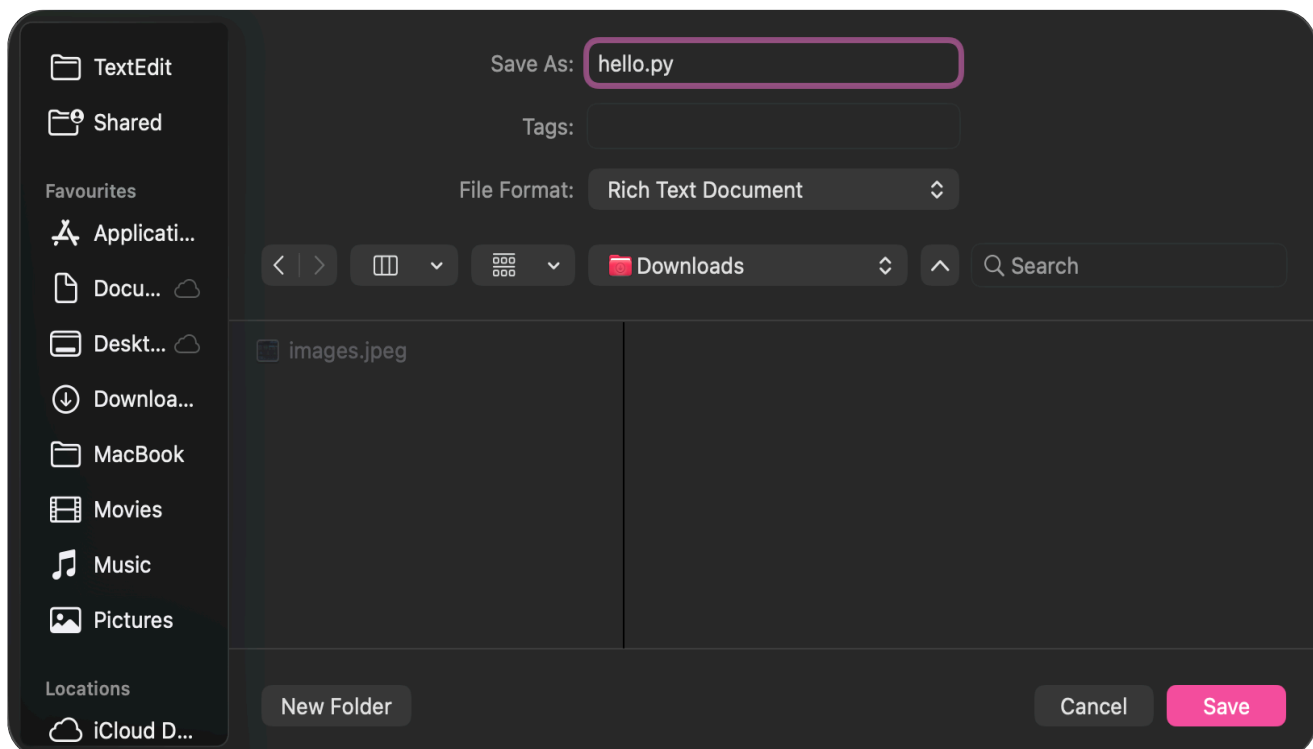
Step 2: Select Save or Save As.

Step 3: Choose the folder where you want to save the program.

Step 4: Enter a suitable file name.

Step 5: Click the Save button.

Python programs are saved with the .py extension.



Running a Python Program

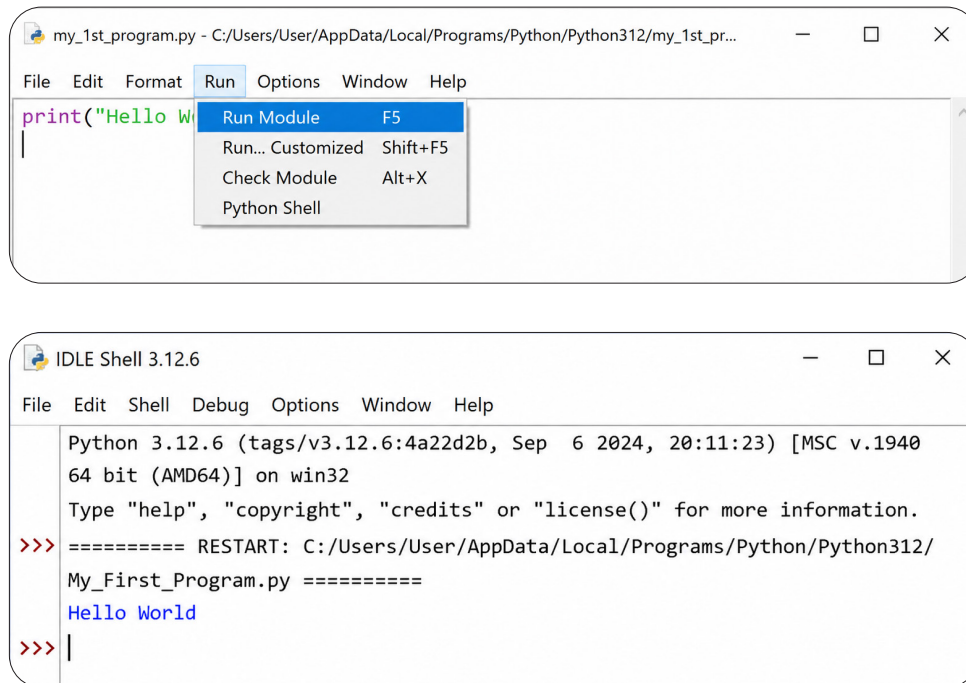
After saving the file, we can execute the program to see its output.

Follow these steps:

Step 1: Click the Run menu.

Step 2: Select Run Module.

Step 3: The program is executed, and the output appears in the Python Shell window.



Input and Output

In Python, special functions are used to receive input from the user and display the output. Learning these functions is one of the first steps in writing useful programs.

The input() Function

The `input()` function is used to accept data from the user while the program is running. It allows the user to enter information such as a name, age, number, or any other value.

When the program reaches the `input()` function, it waits until the user enters a value and presses the Enter key.



```
1 name = input("Enter Your Name: ")
2 >>> Enter Your Name: "Krishna"
```



The print() Function

The print() function is used to display messages, numbers, or the results of calculations on the screen. It is one of the most commonly used functions in Python.

Text or messages to be displayed are written inside quotation marks.



```
1 print("Python is Easy !")
2 print("I Will learn it in 1 Month")
```

Variables in Python

Suppose you want to store your favourite colour, your age, or your marks in a program. Python needs a place in the computer's memory to keep this information. This place is called a variable. A variable is a named memory location used to store data. The value stored in a variable can change while the program is running, which is why it is called a variable.

Think of a variable as a labelled storage box. The label tells us what is stored inside, and the contents can be replaced whenever needed.



```
1 student = "Ananya"
2 age = 11
3 marks = 95
```

Rules for Naming Variables

Python follows certain rules while naming variables.

- A variable name must begin with a letter (A–Z or a–z) or an underscore (_).
- It cannot begin with a number.
- It can contain letters, numbers, and underscores.
- Spaces are not allowed in variable names.
- Special symbols such as @, #, %, and & cannot be used.
- Python keywords such as if, for, and print should not be used as variable names.

Valid Variable Names: name, student_name, marks1, _age

Invalid Variable Names: 2name, student name, total%, class



Data Types

In a Python program, different kinds of information are stored in variables. The kind of value that a variable can store is called its data type. In simple words, a data type tells Python what type of data is being stored in a variable. For example, a student's name is stored as text, while the student's age or marks are stored as numbers.

Choosing the correct data type helps Python process information correctly and perform different operations on it.

- **int:** An integer is a whole number without any decimal point. It can be positive, negative, or zero. **Example:** `a=28, num=18`
- **float:** A float is a number that contains a decimal point. It is used to store values such as height, weight, price, or distance. **Example:** `num=18.45, b=1.8`
- **string:** A string is a collection of one or more characters enclosed within single (') or double (") quotation marks. Strings are used to store names, sentences, addresses, and other text. **Example:** `name="Virat", hobby="Cricket"`

Comments in Python

While writing programs, programmers often include notes that explain what a particular statement or section of the program does. These notes are called comments.

A comment is ignored by Python during program execution. It is written only to make the program easier to read and understand. Comments are especially useful when a program becomes long, as they help both the programmer and others understand the purpose of different parts of the code.

Single Line Comment

A single-line comment is used when the explanation fits on one line. It always begins with the hash (#) symbol.

Example:



```
1 # printing a string
2 print("Hello World")
```

Multiple-Line Comment

Sometimes we need to write a longer explanation that spans more than one line. Such comments are called multi-line comments. In Python, multi-line comments can be written by placing the text inside triple quotation marks (''' ''' or """ """)



Example:



```
1 ''' This is a multiple line comment '''
2 print("Hello World")
```

Operators

In mathematics, we use symbols such as +, −, ×, and ÷ to perform calculations. Similarly, Python uses special symbols called operators to perform different types of operations on data. An operator is a symbol that tells Python to perform a specific operation on one or more values. The values on which an operator works are called operands.

Arithmetic Operators

Arithmetic operators are used to perform mathematical calculations such as addition, subtraction, multiplication, and division. These operators are commonly used while writing Python programs.

Operator	Name	Description	Example (x=7, y=3)	Output
+	Addition	Adds the value on the right to the value on the left.	$x + y$	10
−	Subtraction	Subtracts the right operand from the left operand.	$x - y$	4
*	Multiplication	Multiplies the two operands.	$x * y$	21
/	Division	Divides the left operand by the right operand and returns the result as a decimal number.	x / y	2.33
%	Modulus	Returns the remainder after division.	$x \% y$	1
//	Floor Division	Divides the operands and returns only the whole-number part of the result.	$x // y$	2
**	Exponent	Raises the left operand to the power of the right operand.	$x ** y$	343



Program 1: To perform all Arithmetic Operations

```
1 a = 9
2 b = 4
3
4 print("Addition =", a + b)
5 print("Subtraction =", a - b)
6 print("Multiplication =", a * b)
7 print("Division =", a / b)
8 print("Floor Division =", a // b)
9 print("Modulus =", a % b)
10 print("Exponent =", a ** b)
```

```
Python — -zsh — 170x35

[mac@MacBook-Pro Python % python3 demo.py
Addition = 13
Subtraction = 5
Multiplication = 36
Division = 2.25
Floor Division = 2
Modulus = 1
Exponent = 6561
mac@MacBook-Pro Python %
```

Assignment Operators

Assignment operators are used to assign values to variables. They can also perform an operation and assign the result back to the same variable in a single step. These operators make Python programs shorter, easier to read, and more efficient.

Operator	Name	Description	Example
=	Assignment	Assigns the value on the right side to the variable on the left side.	x = 10
+=	Addition Assignment	Adds the given value to the variable and stores the result in the same variable.	x += 5



--=	Subtraction Assignment	Subtracts the given value from the variable and stores the result.	x -= 4
*=	Multiplication Assignment	Multiplies the variable by the given value and stores the result.	x *= 2
/=	Division Assignment	Divides the variable by the given value and stores the result.	x /= 2
%=	Modulus Assignment	Finds the remainder and stores it in the same variable.	x %= 4
//=	Floor Division Assignment	Performs floor division and stores the whole-number result.	x //= 3
**=	Exponent Assignment	Raises the variable to the given power and stores the result.	x **= 2

Program 2: To perform all Assignment Operations

```

1 x = 10
2
3 x += 5
4 print("After += :", x)
5 x -= 3
6 print("After -= :", x)
7 x *= 2
8 print("After *= :", x)
9 x /= 4
10 print("After /= :", x)
11 x = 10
12 x %= 3
13 print("After %= :", x)

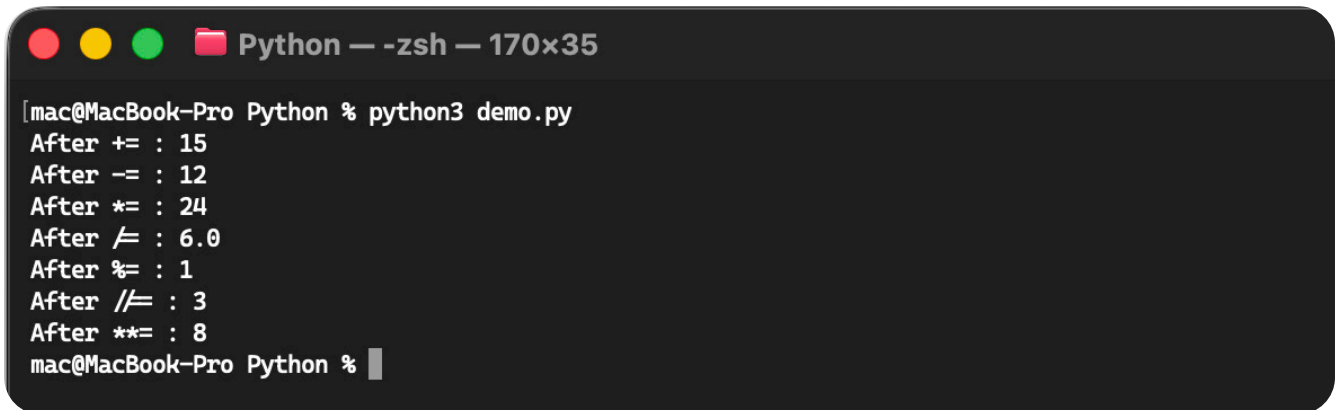
```



```

14 x = 10
15 x //= 3
16 print("After //= :", x)
17 x = 2
18 x **= 3
19 print("After **= :", x)

```



```

Python — -zsh — 170x35

[mac@MacBook-Pro Python % python3 demo.py
After += : 15
After -= : 12
After *= : 24
After /= : 6.0
After %= : 1
After //= : 3
After **= : 8
mac@MacBook-Pro Python %

```

Logical Operators

Sometimes, a program needs to check more than one condition before making a decision. For example, a student may receive a prize only if they score more than 90 marks and have good attendance. In such situations, logical operators are used. Logical operators combine two or more conditions and return either True or False depending on whether the conditions are satisfied.

Operator	Name	Description	Example (a=true, b=false)	Output
and	Logical AND	Returns True only when both conditions are True. If any one condition is False, the result is False.	a and b	False
or	Logical OR	Returns True if at least one condition is True. It returns False only when both conditions are False.	a or b	True
not	Logical NOT	Reverses the result of a condition. True becomes False and False becomes True.	a not b	False



Program 3: To perform all Logical Operations

```
1 is_member = True
2 has_ticket = False
3
4 print(is_member and has_ticket)
5 print(is_member or has_ticket)
6 print(not is_member)
7 print(not has_ticket)
```

```
Python — -zsh — 170x35
[mac@MacBook-Pro Python % python3 demo.py
False
True
False
True
mac@MacBook-Pro Python %
```

Relational Operators

Sometimes, a program needs to check more than one condition before making a decision. For example, a student may receive a prize only if they score more than 90 marks and have good attendance. In such situations, logical operators are used. Logical operators combine two or more conditions and return either True or False depending on whether the conditions are satisfied.

Operator	Name	Description	Example (x=12, y=8)	Output
==	Equal to	Checks whether both values are equal.	x == y	False
!=	Not Equal to	Checks whether the two values are not equal.	x != y	True
>	Greater Than	Checks whether the left value is greater than the right value.	x > y	True



<	Less Than	Checks whether the left value is smaller than the right value.	$x < y$	False
>=	Greater Than or Equal To	Checks whether the left value is greater than or equal to the right value.	$x \geq y$	True
<=	Less Than or Equal To	Checks whether the left value is less than or equal to the right value.	$x \leq y$	False

Program 4: To perform all Relational Operations



```

1 first = 18
2 second = 25
3
4 print(first > second)
5 print(first < second)
6 print(first == second)
7 print(first != second)
8 print(first >= second)
9 print(first <= second)

```



Python — -zsh — 170x35

```

[mac@MacBook-Pro Python % python3 demo.py
False
True
False
True
False
True
mac@MacBook-Pro Python %

```



Precedence of Operators

Sometimes, a Python statement contains more than one operator. In such cases, Python follows a fixed order to decide which operation should be performed first. This order is known as the precedence of operators. Understanding operator precedence helps us predict the correct output of a program and avoid mistakes while solving mathematical expressions.

The highest precedence is at the top.

Operator	Name
()	Parenthesis
**	Exponent
*, /, %, //	Multiplication, Division, Floor Division and Modulus
+, -	Addition and Subtraction
==, !=, >, <, <=, >=	Relational Operators
=, +=, +=, *=, /=, %=, **=, //=	Assignment Operators
and, or, not	Logical Operators

Some more Programs

Program 5: To calculate the Area of a Rectangle



```
1 length = float(input("Enter the length: "))
2 breadth = float(input("Enter the breadth: "))
3
4 area = length * breadth
5 print("Area of the rectangle is:", area)
```



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```
[mac@MacBook-Pro Python % python3 demo.py
Enter the length: 24
Enter the breadth: 12
Area of the rectangle is: 288.0]
```



Program 6: To calculate Average Marks of four subjects



```
1 english = int(input("Enter English marks: "))
2 maths = int(input("Enter Maths marks: "))
3 science = int(input("Enter Science marks: "))
4 computer = int(input("Enter Computer marks: "))
5
6 average = (english + maths + science + computer) / 4
7
8 print("Average Marks =", average)
```

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```
[mac@MacBook-Pro Python % python3 demo.py
Enter English marks: 78
Enter Maths marks: 98
Enter Science marks: 75
Enter Computer marks: 88
Average Marks = 84.75
mac@MacBook-Pro Python %
```

Program 7: To calculate Simple Interest



```
1 principal = float(input("Enter the principal amount: "))
2 rate = float(input("Enter the rate of interest: "))
3 time = float(input("Enter the time (in years): "))
4
5 interest = (principal * rate * time) / 100
6
7 print("Simple Interest =", interest)
```



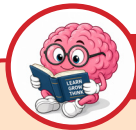
Python — -zsh — 170x35

```
[mac@MacBook-Pro Python % python3 demo.py
Enter the principal amount: 50000
Enter the rate of interest: 12
Enter the time (in years): 10
Simple Interest = 60000.0
mac@MacBook-Pro Python %
```



Did You Know?

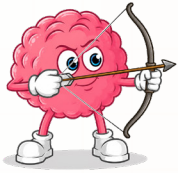
Python is one of the most popular programming languages in the world and is widely used in Artificial Intelligence (AI), Machine Learning, Robotics, Web Development, Game Development, and Data Science.



Quick Recap

- A program is a set of instructions given to a computer to perform a task.
- Python is a simple and popular high-level programming language.
- Python programs can be written in Interactive Mode and Script Mode.
- The `input()` function accepts data from the user, while the `print()` function displays the output.
- A variable is used to store data in a computer's memory.
- Python commonly uses Integer (`int`), Float (`float`), and String (`str`) data types.
- Comments help explain a program and are ignored during execution.
- Operators are special symbols used to perform calculations and comparisons.
- Python supports Arithmetic, Assignment, Logical, and Relational operators.
- Operator precedence determines the order in which operations are performed in an expression.





TEST YOUR SKILLS



A. Choose the correct option

- What will be the output of the following expression ($18 // 4$)?
a. 4.5 ☐ b. 4 ☐
c. 2 ☐ d. 5 ☐
- Which of the following is a valid variable name in Python?
a. 2marks ☐ b. student_name ☐
c. student-name ☐ d. student name ☐
- Which of the following operator is used to find the remainder after division?
a. / ☐ b. // ☐
c. % ☐ d. * ☐
- Which of the following statement about Python comments is correct?
a. They are executed by Python. ☐ b. They improve readability ☐
c. Always produce output ☐ d. Used to store variables ☐
- Which operator has the highest precedence?
a. + ☐ b. * ☐
c. () ☐ d. == ☐

B. State True or False

- Python is an interpreted programming language. _____
- Variable names can begin with a number. _____
- The input() function is used to receive data from the user. _____
- The % operator returns the remainder after division. _____
- Comments are executed as part of the program. _____

C. Fill in the blanks

- Python was developed by _____.
- The _____ operator is used to assign a value to a variable.
- A number with a decimal point is stored as a _____ data type.
- The _____ function is used to display output on the screen.
- _____ make programs easier to understand.



D. Short Answer Questions

1. What is a programming language?
2. What is the purpose of the print() function in Python?
3. What is a variable? Give one example.
4. What are data types?
5. What is operator precedence?

E. Long Answer Questions

1. Explain the different data types used in Python with suitable examples.
2. What are comments? Explain the types of comments used in Python with examples.
3. Explain the two programming modes available in Python IDLE.
4. Define the following:
 - Arithmetic Operators
 - Assignment Operators
 - Logical Operators
 - if-elif-else ladder

F. Competency-Based Questions

- a) Riya wants to write a Python program to display her name and age on the screen. Which Python function should she use to display the information? Write a simple program for her.
- b) Aman is creating a program to calculate the total marks of five subjects. He wants to add comments so that anyone reading the program can understand its purpose. Why are comments useful? Write a suitable multi-line comment for his program.



Activity

Write Python Programs to:

1. Calculate the perimeter of a rectangle.
2. Find the average of three numbers entered by the user.
3. Input the name and age of a student and display them.
4. Calculate the area of a circle.
5. Find the square of a number entered by the user.





Practical Exercises

1. Predict the output:

a.



```
1 num = 16
2 print(num // 3)
```

b.



```
1 value = 12
2 value += 8
3 print(value)
```

c.



```
1 a = 15
2 b = 5
3 print(a + b)
```

d.



```
1 name = "Rohan"
2 print(name)
```

Teacher's Note

- Encourage students to write and execute simple Python programs in both Interactive Mode and Script Mode.
- Discuss the importance of using meaningful variable names and comments while writing programs.
- Provide students with simple real-life problems that can be solved using Python programs.
- Allow students to practise arithmetic, relational, and logical operators through classroom activities.



Intelligence & AI Approaches

Learning Goals

- Intelligence
- Types of Intelligence
- AI Approach
- Types of AI Approach



Let's Warm Up the Brain

Write an eight-letter word that starts with "R" and ends with "G".

R								G
---	--	--	--	--	--	--	--	---

Clues:

- It is the ability to think logically.
- It helps us make good decisions.
- It is an important part of intelligence.

Have you ever wondered why some people solve maths problems quickly, while others are good at drawing, singing, or playing sports? Every person is intelligent in a unique way. Intelligence is the ability to learn, think, understand, solve problems, and make decisions.

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that usually require human intelligence. AI helps us in many ways, such as using voice assistants, recognising faces, translating languages, and recommending videos or songs.

In this chapter, you will learn about different types of intelligence and understand how AI works using rule-based and learning-based approaches.

Intelligence

Intelligence is the ability to understand, learn, reason, solve problems, and apply knowledge in different situations. It helps us make decisions, recognise patterns, remember information, and adapt to new experiences.

Every person has different strengths. Some are good at mathematics, while others are talented in music, art, sports, or communication.

According to psychologist Howard Gardner, intelligence is not limited to one ability. He proposed that humans possess different kinds of intelligence.

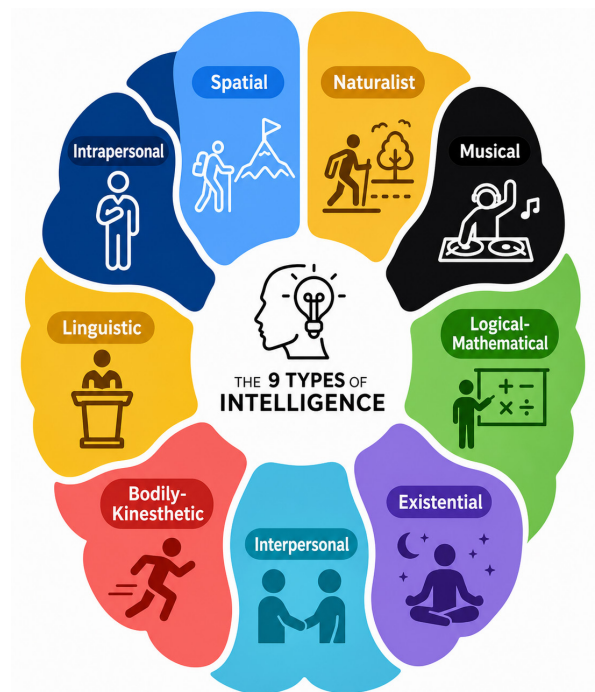
Types of Intelligence

Human beings have the ability to learn, understand, reason, communicate and solve problems. However, every person is unique and may excel in different areas. According to the theory of Multiple Intelligences proposed by psychologist Howard Gardner, intelligence is not just about academic performance. It includes a variety of abilities such as logical thinking, communication, creativity, music, physical coordination, understanding nature and interacting with others.



The **figure** shows the nine types of intelligence identified by Howard Gardner. These intelligences help us recognise our strengths and understand that every individual has unique abilities. A person may have more than one type of intelligence, and each type can be developed through learning, practice and experience.

Understanding the different types of intelligence helps us appreciate our own talents as well as the abilities of others. It also encourages us to learn in ways that suit our strengths while improving areas where we need more practice. Schools, families and society can support every learner by recognising and nurturing these different forms of intelligence.



Visual-Spatial Intelligence

People with Visual-Spatial Intelligence can understand pictures, shapes, colours, maps and diagrams easily. They can imagine objects in their minds and think creatively through visuals.

These people possess the following qualities:

- Enjoy drawing, painting and designing.
- Can read maps and charts easily.
- Recognise colours, shapes and patterns quickly.
- Imagine objects from different angles.
- Like solving jigsaw puzzles and mazes.

Examples: Architects, painters, photographers, interior designers and graphic designers.

Verbal-Linguistic Intelligence

People with Verbal-Linguistic Intelligence are good at using words effectively. They enjoy reading, writing, speaking and expressing their thoughts clearly.

These people possess the following qualities:

- Love reading books and stories.
- Enjoy writing essays, poems or stories.
- Speak confidently in front of others.
- Imagine objects from different angles.
- Like solving jigsaw puzzles and mazes.
- Notice small visual details around them.

Examples: Teachers, writers, journalists, lawyers and news presenters.



Logical-Mathematical Intelligence

People with Logical-Mathematical Intelligence enjoy reasoning, analysing situations and solving problems. They like working with numbers, patterns and logical thinking.

These people possess the following qualities:

- Solve mathematical problems quickly.
- Think logically before making decisions.
- Like conducting simple experiments.
- Enjoy puzzles, riddles and brain teasers.
- Recognise patterns and relationships.
- Enjoy coding and problem-solving activities.

Examples: Scientists, engineers, programmers, accountants and mathematicians.

Bodily-Kinesthetic Intelligence

People with Bodily-Kinesthetic Intelligence use their body skilfully to perform different activities. They learn better by doing practical work and physical movement.

These people possess the following qualities:

- Enjoy sports and outdoor games.
- Learn by practising rather than reading.
- Handle tools and equipment carefully.
- Have good hand-eye coordination.
- Perform dance, yoga or gymnastics well.
- Like activities that require movement.

Examples: Athletes, dancers, surgeons, actors and craftsmen.

Musical Intelligence

People with Musical Intelligence understand rhythm, melody and sound. They enjoy singing, listening to music and playing musical instruments.

These people possess the following qualities:

- Enjoy singing and musical instruments.
- Remember songs easily.
- Notice different sounds around them..
- Recognise musical notes and rhythms.
- Create or compose simple tunes.
- Express emotions through music.

Examples: Singers, musicians, composers and music teachers.

Interpersonal Intelligence

People with Interpersonal Intelligence understand the feelings and emotions of others. They communicate well and enjoy working with people.

These people possess the following qualities:

- Work well in teams.
- Listen carefully to others.
- Maintain positive relationships.
- Communicate ideas clearly.
- Help friends solve problems.
- Show leadership qualities.

Examples: Teachers, counsellors, managers, leaders and social workers.

Intrapersonal Intelligence

People with Intrapersonal Intelligence understand themselves well. They know their strengths, weaknesses and emotions, and like setting personal goals.

These people possess the following qualities:

- Understand their own feelings.
- Know their strengths and weaknesses.
- Think carefully before making decisions.
- Set personal goals.
- Learn from their experiences.
- Enjoy self-reflection and independent learning.

Examples: Philosophers, psychologists, writers and researchers.

Naturalistic Intelligence

People with Naturalistic Intelligence love nature and enjoy observing plants, animals and the environment. They care about protecting natural resources.

These people possess the following qualities:

- Love plants and animals.
- Observe nature carefully.
- Recognise different species of plants and animals.
- Enjoy gardening and farming.
- Like exploring forests, parks and wildlife.

Examples: Farmers, botanists, zoologists, environmentalists and wildlife photographers.

Existential Intelligence

People with Existential Intelligence think deeply about life, nature and the universe. They enjoy asking meaningful questions and understanding the world around them.

These people possess the following qualities:

- Think about the purpose of life.
- Look at situations from different angles.
- Reflect on their thoughts and beliefs.
- Ask thoughtful questions.
- Connect ideas with real-life experiences.
- Enjoy space, nature and the universe.

Examples: Philosophers, spiritual leaders, scientists and thinkers.

AI Approach

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, recognising patterns and making decisions. AI systems are designed to analyse information, solve problems and improve their performance over time. Today, AI is used in many areas, including education, healthcare, transportation, banking and entertainment, making our daily lives easier and more efficient. To perform different tasks, AI follows various methods known as AI approaches.

The most common types of AI Approaches are: **Rule Based** and **Learning Based** Approach.



Rule Based Approach

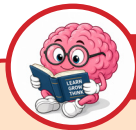
A Rule-Based Approach is one of the simplest methods used in Artificial Intelligence. In this approach, the computer follows a fixed set of rules or instructions provided by humans. These rules are usually written in the form of IF-THEN statements. The AI system cannot think or learn on its own; it simply follows the programmed rules to make decisions. If a situation is not covered by the given rules, the system cannot provide the correct response.

Examples: Calculator, traffic light controller, password validation system and automatic ticket vending machine.

Learning Based Approach

A Learning-Based Approach enables an AI system to learn from data and past experiences instead of relying only on fixed rules. The system analyses large amounts of information, recognises patterns and improves its performance over time. As it receives more data, it becomes more accurate and can make better decisions without being explicitly programmed for every situation. This approach is widely used in modern AI applications.

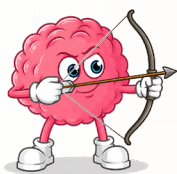
Examples: Face recognition, voice assistants, ChatGPT, recommendation systems and spam email filters.



Quick Recap

- *Intelligence helps us think, learn, and solve problems.*
- *Howard Gardner proposed multiple types of intelligence.*
- *Everyone has different strengths and abilities.*
- *AI allows machines to perform intelligent tasks..*
- *AI mainly works using Rule-Based and Learning-Based approaches.*





TEST YOUR SKILLS



A. Choose the correct option

- Which of the following is an example of Artificial Intelligence?
a. Calculator () b. USB Drive ()
c. Keyboard () d. Voice Assistant ()
- Which AI approach learns from data and improves its performance over time?
a. Rule-Based Approach () b. Learning-Based Approach ()
c. Manual Approach () d. None of these ()
- Which of the following is an example of a virtual assistant?
a. Alexa () b. MS Paint ()
c. Calculator () d. Notepad ()
- Who proposed the theory of Multiple Intelligences?
a. Alan Turing () b. Howard Gardner ()
c. Charles Babbage () d. John von Neumann ()
- A rule-based AI system mainly works by following:
a. Random guesses () b. Predefined rules ()
c. Human emotions () d. Pictures ()

B. State True or False

- Every person has the same type of intelligence. _____
- Learning-based AI can improve with experience. _____
- A rule-based AI system can learn by itself without any instructions. _____
- Musical intelligence is related to rhythm and sounds. _____
- AI is used only in robots. _____

C. Fill in the blanks

- _____ proposed the theory of Multiple Intelligences.
- AI systems using fixed instructions follow _____.
- A _____ approach improves by learning from data.
- People who enjoy gardening and observing wildlife possess _____ intelligence.
- _____ is the ability to think, learn and solve problems.



D. Short Answer Questions

1. What is intelligence?
2. Name any four types of intelligence.
3. What is Artificial Intelligence?
4. What is a Rule-Based Approach?
5. What is a Learning-Based Approach?

E. Long Answer Questions

1. Explain the Logical-Mathematical Intelligence with suitable examples.
2. Describe any four types of intelligence proposed by Howard Gardner.
3. Differentiate between Rule-Based and Learning-Based AI approaches.
4. Explain how Artificial Intelligence is useful in our daily life.

F. Competency-Based Questions

- a) Ayush enjoys solving Sudoku puzzles, coding simple programs and conducting science experiments. Which type of intelligence does he have? Give one reason.
- b) Sneha loves singing, plays the keyboard and quickly remembers melodies. Which type of intelligence is she showing?
- c) Krishna always understands his friends' feelings and helps settle disagreements in class. Which intelligence does he possess?



Explore

Search the Internet and find three AI applications that help students learn better. Write their names and one use of each.



Activity

1. Identify the Pattern

Find the next number.

3, 6, 9, 12, 15, _____

Which type of intelligence is mainly used here? _____

2. Crack the Code

Unscramble the word:

LARITAFICI _____





Practical Exercises

1. Create a colourful chart showing the nine types of intelligence with one example of each.
2. Prepare a presentation explaining the difference between Rule-Based AI and Learning-Based AI using real-life examples.



SDG Activity

Artificial Intelligence is making education smarter and more accessible. Find out how AI-powered learning apps help students improve their reading, writing or mathematics skills. Prepare a short report with pictures and present it in your class.

Teacher's Note

- *Encourage students to identify their own strengths by discussing the different types of intelligence.*
- *Organise simple classroom activities such as storytelling, puzzles, drawing, music or teamwork to help students recognise different intelligences.*

